Functional Analysis

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1 Metric Spaces.

1.1 Topological Spaces

Definition 1.1. Given a set X, a topology τ on X is a subset of $\mathcal{P}(X)$ s.t.

1. $\varnothing, X \in \tau$
2. Closed under union: $E_\alpha \in \tau \; \forall \alpha \in I \implies \bigcup_{\alpha \in I} E_\alpha \in \tau$
3. Closed under finite intersection: $E_1, E_2, \cdots, E_n \in \tau \implies \bigcap_{i=1}^{n} E_i \in \tau$

We say that (X, τ) is a topological space.

Definition 1.2. Let (X, τ) a topological space and $A \subseteq X$

- A open $\iff A \in \tau$.
- A is closed $\iff X \setminus A$ is open.

Definition 1.3. (X, τ) topological space and $x \in X$. A neighborhood of x is an open set $U_x \ni x$.

Definition 1.4. (X, τ) topological space, ,and $A \subseteq X$. Then x is an interior point of $A \iff \exists U_x$ an open neighborhood of x s.t. $U_x \subseteq X$.

We denote by $\mathring{A}$ the set of interior points of A. Notice $\mathring{A} \subseteq A$.

Now x is a limit point of $A \iff \forall U_x$ a neighborhood of x, we have that $U_x \setminus \{x\} \cap A \neq \varnothing$. A' denotes the set of all limit points of A and $\overline{A} := A \cap A'$ denotes the closure of A.

Proposition 1.5. *1. $\mathring{A}$ is an open set. In fact, A is open $\iff A = \mathring{A}$.*

2. $\overline{A}$ is closed. In fact, A is closed $\iff A' \subseteq A \iff \overline{A} = A$.

Consequence, A is open iff $\forall x \in A,\ \exists U_x \ni x$ an open neighborhood s.t. $U_x \subseteq A$.

Proof. Easy. □

Definition 1.6. Let $(X, \tau_x),\ (Y, \tau_y)$ topological spaces, and $f\colon X \to Y$ and $x_0 \in X$. Then f is continuous at $x_0 \iff \forall V$ neighborhood of $f(x_0)$ there exists a neighborhood U of x_0 s.t. $f(U) \subseteq V$.

f is said to be continuous on X iff f is continuous at every $x_0 \in X$.

Proposition 1.7. *$f\colon X \to Y$ continuous iff $f^{-1}(G)$ is open in X for every G open in Y.*

Definition 1.8. (X, τ) topological spaces, and x_n a sequence in X. We say that x_n converges to $x_0 \in X$ iff for every U_{x_0} neighborhood of x_0, $\exists N \in \mathbb{N}$ s.t. $x_n \in U_{x_0}\ \forall n \geq N$.

Example 1.9. $x_n = \frac{1}{n}$.

- In $(\mathbb{R}, \tau_{\text{trivial}})$, this sequence converges to every point in $\mathbb{R}$.
- In $(\mathbb{R}, \tau_{\text{disc}})$, this sequence doesn't converge to any point (we say that it diverges).

Proposition 1.10. *(X, τ) topological space. $A \subseteq X$, x_n sequence in A that converges to x_0. Then $x_0 \in \overline{A}$.*

Proposition 1.11. *$f\colon (X, \tau_x) \to (Y, \tau_y)$ and $x_0 \in X$ with f continuous at x_0. If x_n converges to x_0, then $f(x_n)$ converges to $f(x_0)$.*

Definition 1.12. (X, τ) topological space, $A \subseteq X$. We say that A is dense in X iff $\overline{A} = X$. We say that X is separable iff it has a countable dense subset.

Example 1.13. $(\mathbb{R}^n, |\cdot|)$ is separable. $(\mathbb{R}, \tau_{\text{disc}})$ is not separable. $(X, \tau_{\text{trivial}})$ is always separable.

Definition 1.14. (X, τ) is a Hausdorff space iff $\forall x, y \in X\ x \neq y\ \exists U_x, U_y$ neighborhoods of x, y respectively s.t. $U_x \cap U_y = \varnothing$.

Proposition 1.15. *The limit of a convergent sequence in a Hausdorff space is unique.*

Proof. Assume x_n converges to x, y with $x \neq y$, since Hausdorff we have U_x, U_y neighborhoods with $U_x \cap U_y = \varnothing$. Now for n big enough $x_n \in U_x$ for $n \geq N$ then $x_n \notin U_y$ for these $n \geq N$, contradiction. □

Definition 1.16. X a set, then a metric on X is a function $d\colon X \times X \to \mathbb{R}$ s.t.

1. $d(x,y) \geq 0,\ \forall x, y \in X$ with equality iff $x = y$.
2. $d(x,y) = d(y,x)\ \forall x, y \in X$.
3. $d(x,z) \leq d(x,y) + d(y,z)\ \forall x, y, z \in X$

We say that (X, d) is a metric space.

Definition 1.17. $x, \in X$ and $r \geq 0$ then

$$B(x,r) = \{y \in X\colon d(x,y) < r\}$$

Now define

$$\tau_d\,\{U \in \tau_d \iff \forall x \in U \exists r \geq 0 s.t. B(x,r) \subseteq U\}$$

Proposition 1.18. *the $B(x,r)$. τ_d is a topology on X, called the metric topology. Then (X, τ_d) is always Hausdorff and τ_d is the smallest topology containing all*

Proposition 1.19. *Take (X,d) metric space and x_n converges to x_0 iff $\forall \varepsilon > 0,\ \exists N \in \mathbb{N}$ s.t. $d(x_n, x_0) < \varepsilon\ \forall n \geq N$.*

Proposition 1.20. *(X,d) metric space and $E \subseteq X$, then $x \in \overline{E}$ iff $\exists x_n \in E$ a convergent sequence with $x_n \to x$*

Proof. ($\Longleftarrow$) Any topological space.
($\Longrightarrow$) Take $x \in \overline{E} = E \cup E'$, if $x \in E$ take $x_n = x$. Now if $x \in E'$, then for every $n \in \mathbb{N},\ \exists x_n \in B^*(x, 1/n) \cap E$ then $d(x_n, x) < 1/n$, by Archimedean property we get that $x_n \to x$.

□

Proposition 1.21. *(X,d) metric space with $E \subseteq X$, then E is closed iff for every convergent sequence $x_n \in E$ the limit as $n \to \infty$ of $x_n \in E$.*

Definition 1.22. (X,d) metric space, then $E \subseteq X$ is bounded iff $E \subseteq B(x_0, R)$ for some $x_0 \in x$ and $R > 0$. In particular, a sequence x_n is bounded iff $\exists M$ s.t. $|x_n| \leq M\ \forall n \in \mathbb{N}$.

Example 1.23 (Some examples of some separable or not separable metric spaces)**.**

- $(\mathbb{R}, |\cdot|)$ separable with countable dense set $\mathbb{Q}^n$.
- The space $C([a,b])$ of real valued continuous functions with $d_\infty(f,g) = \max_{[a,b]} |f(x) - g(x)|$. By Stone-Weierstrass, for every $f \in C([a,b])$ there exists a sequence of polynomials P_n s.t. $\lim_{n\to\infty} P_n = f$ uniformly. Let's call $P(\mathbb{R})$ the space of polynomials and fix $\varepsilon > 0$ and $f \in C([a,b])$. Then $\exists P \in P(\mathbb{R})$ s.t. $d_\infty(P, f) < \varepsilon/2$ with

$$p(x) = c_k x^k + c_{k-1} x^{k-1} + \cdots + c_1 x + c_0$$

Now for each $c_i \in q_i \in \mathbb{Q}$ s.t. $|c_i - q_i| < \frac{\varepsilon}{2M}$ where $M \geq \sum_{i=0}^{k} |x|^i, \forall x \in [a,b]$. Now let

$$q(x) = q_k x^k + \cdots + q_1 x + q_0$$

So we get that

$$\begin{aligned} |p(x) - q(x)| &\leq \sum_{i=0}^{k} \sum_{q_i - c_i} |x^i| \\ &\leq \frac{\varepsilon}{2M} \sum_{i=0}^{k} |x|^i \end{aligned}$$

So we get that

$$d_\infty(q, f) < \varepsilon$$

Hence $C([a,b])$ is separable.

- Let $\ell^\infty(\mathbb{R})$ be the space of bounded sequences. Now define

$$x = (x_1, x_2, \cdots), \quad y = (y_1, y_2, y_3, \cdots)$$

and

$$d_\infty(x, y) = \sup_{i \in \mathbb{N}} |x_i - y_i|$$

Assume it's separable, let M be the countable dense subset. Let $A = \{0,1\}^{\mathbb{N}}$ be the space of all sequences with entries 0 or 1, clearly A is uncountable. But if $x, y \in A$ with $x \neq y$ then $d_\infty(x,y) = 1$. For every $a \in A$, construct $B(a, 1/2)$, let m_a be an element of $B(a,1/2) \cap M$ (uses AOC and density of m_a). Clearly if $a \neq b$, we have $B(a,1/2) \cap B(b,1/2) = \varnothing$, then $a \to m_a$ is injective, hence M is uncountable. Contradiction.

Definition 1.24. Let (X,d) a metric space, given a sequence x_n in X, we say that x_n is Cauchy iff $\forall \varepsilon > 0,\ \exists N$ s.t.

$$d(x_n, x_m) < \varepsilon,\ \forall n, m \geq N$$

Proposition 1.25. *Every Cauchy sequence is bounded, every convergent sequence is Cauchy.*

Definition 1.26. (X,d) is complete iff every Cauchy Sequence converges.

Example 1.27. $(\mathbb{R}^n, |\cdot|)$ is complete.

Some properties of metric spaces.

i) if x_n converges, then x_n is bounded.

ii) Let C be the space of convergent sequences, then $C \subseteq \ell^\infty$. We want to show that C is closed in ℓ^∞. Let $x^1 = (x_{1,1}, x_{1,2}, \cdots), x^2 = (x_{2,1}, x_{2,2}, \cdots), x^3 = (x_{3,1}, x_{3,2}, \cdots), \cdots$ be a sequence in C that converges to some sequence $x = (x_1, x_2, \cdots) \in \ell^\infty$. Let $\varepsilon > 0$, since x^n converges to x in ℓ^∞, $\exists N$ s.t $d_\infty(x^n, x) < \varepsilon/3\ \forall n \geq N$. In particular for $n = N$ we have

$$|x_{N,i} - x_i| < \varepsilon/3, \quad \forall i = 1, 2, 3, \cdots$$

Now x^N is Cauchy, then $\exists i_0$ s.t.

$$|x_{N,i} - x_{N,j}| < \varepsilon/3, \quad \forall i, j \geq i_0$$

Hence

$$\begin{aligned} |x_i - x_j| &\leq |x_i - x_{N,i}| + |x_{N,i} - x_{N,j}| + |x_{N,j} - x_j| \\ &< \varepsilon \end{aligned}$$

Then the sequence x is Cauchy, since $\mathbb{R}$ is complete we have that $x \in C$. So C is closed.

iii) Is $\ell^\infty(\mathbb{R})$ complete. Let $x^1 = (x_{1,1}, x_{2,1}, \cdots), x^2 = (x_{2,1}, x_{2,2}, \cdots), \cdots$ be a Cauchy sequence in $\ell^\infty(\mathbb{R})$. We define

$$x_i := (x_i^1, x_i^2, x_i^3, \cdots)$$

Clearly we have that for each $i \in \mathbb{N}$,

$$d_\infty(x^k, x^l) \geq |x_{k,i} - x_{l,i}|$$

Hence it must be the case that all the x_i's are Cauchy, and hence converge to some x_i^* (since $\mathbb{R}$ is complete). Define

$$x^* := (x_1^*, x_2^*, \cdots)$$

Now by Cauchy we know that $\exists N \in \mathbb{N}$ s.t.

$$d_\infty(x_n, x_m) < \varepsilon/2, \quad \forall n, m \geq N$$

Therefore $\forall i \in \mathbb{N}$ we have that

$$|x_i^n - x_i^m| < \varepsilon/2$$

But we know that $(x_i^m)_i \to x_i^*$, by continuity of $|\cdot|$, we have

$$|x_i^n - x_i^*| \leq \varepsilon/2, \quad \forall i \in \mathbb{N},\ \forall n \geq N$$

Note that N is fixed over all i's hence

$$d_\infty(x^n, x^*) = \sup_{i \in \mathbb{N}} |x_i^n - x_i| \leq \varepsilon/2 < \varepsilon$$

Hence $\ell^\infty(\mathbb{R})$ is complete.

Note that x^* is bounded because each x^n is bounded in ℓ^∞, therefore $|x_{n,i}| \leq M,\ \forall n, i$, taking $n \to \infty$ we get that $|x_i| \leq M,\ \forall i$.

iv) Is $C([a,b])$ complete. This is just uniformly Cauchy iff uniformly convergent and the uniform limit of continuous functions is continuous.

v) Is $P([a,b])$ closed in $C([a,b])$. No, this is because $P([a,b])$ is dense with nonempty complement. So if its complement was open, it should contain a nonempty open set, contradicting

density of the polynomials.

Proposition 1.28. *Let (X, d_x), (Y, d_y) metric spaces and $f : X \to Y$ and $x_0 \in X$ then f is continuous at x_0 iff $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.*

$$d_y(f(x), f(x_0)) < \varepsilon, \quad x \text{ s.t. } d_x(x, x_0) < \delta$$

iff for every sequence $x_n \to x_0$ we have $f(x_n) \to f(x_0)$.

Proposition 1.29. *(X, d) metric space, $E \subseteq X$. If E is complete, then E is closed.*

Proof. Let x_n be a sequence in E that converges to some $x \in X$, but x_n is Cauchy in E so it converges to $x \in E$ (by uniqueness of the limit). □

Example 1.30. Note that the converse is false, $(\mathbb{Q}, |\cdot|)$ is closed in itself but is not Cauchy.

Proposition 1.31. *Every closed subset of a complete metric space is complete.*

Proof. Take (X, d) complete and $E \subseteq X$ closed. Take a Cauchy sequence $x_n \in E$, since $E \subseteq X$ it converges to some $x \in X$. Now E is closed so we must have $x \in E$. □

Example 1.32. $C \subseteq \ell^\infty(\mathbb{R})$ is complete.

Remark 1.33. (X, d) metric space:

$$|d(x, y) - d(x', y')| \leq d(x, x') + d(y, y')$$

This is because

$$\begin{aligned} d(x, y) &\leq d(x, x') + d(x', y') + d(y', y) \\ d(x, y) - d(x', y') &\leq d(x, x') + d(y, y') \end{aligned}$$

Similarly,

$$d(x', y) - d(x, y) \leq d(x', x) + d(y', y)$$

So we get the desired inequality.

Corollary 1.34. *(X, d) metric space and x_n, y_n Cauchy sequences, then $d(x_n, y_n)$ converge in $\mathbb{R}$.*

Proof. Take $d_n = d(x_n, y_n) \in \mathbb{R}$, we show that d_n is Cauchy. Now

$$\begin{aligned} |d_n - d_m| &= |d(x_n, y_n) - d(x_m, y_m)| \\ &\leq d(x_n, x_m) + d(y_n, y_m) \end{aligned}$$

Now since x_n, y_n are Cauchy then d_n is Cauchy in $\mathbb{R}$. Which by completeness gives that d_n converges. □

Lemma 1.35. *(X, d) metric space, and M a dense subset of X s.t. every Cauchy sequence in M converges in X. Then X is complete.*

Proof. Let x_n be a Cauchy sequence in X, for every $n \in \mathbb{N}$, $\exists y_n \in M$ s.t.

$$d(x_n, y_n) < \frac{1}{n}$$

Let $\varepsilon > 0$, now

$$\begin{aligned} d(y_n, y_m) &\leq d(y_n, x_n) + d(x_n, x_m) + d(x_m, y_m) \\ &< \frac{1}{n} + d(x_n, x_m) + \frac{1}{m} \end{aligned}$$

Now $\exists N$ s.t. $n, m \geq N$ we have

$$d(y_n, y_m) < \varepsilon$$

This is because

$$1/m, 1/n < \varepsilon/3, \quad d(x_n, x_m) < \varepsilon/3$$

Now $y_n \in M$ is Cauchy, so y_n converges to some $y_0 \in X$. Now let ε

$$\begin{aligned} d(x_n, y_0) &\leq d(x_n, y_n) + d(y_n, y_0) \\ &< \frac{1}{n} + d(y_n, y_0) \qquad\qquad\qquad\qquad < \varepsilon \end{aligned}$$

for $n \geq N$ for some large enough index N. Then $x_n \to y_0$ so X is complete. □

Definition 1.36. $\varphi \colon (X, d_X) \to (Y, d_y)$ is an isometry iff

$$d_Y(\varphi(x), \varphi(y)) = d_X(x, y)$$

We say that A is isometric to B iff there exists a bijective isometry $\varphi \colon A \to B$.

Theorem 1.37 (Completeness theorem)**.** *(X, d) metric space, then there exists a complete metric space $(\hat{X}, \hat{d})$ and a dense subset M of $\hat{X}$ s.t. M is isometric to X.*

In fact, the completion is "unique" (up to isometry), we call it the completion of X.

Proof. Let C_X be the space of Cauchy sequences in X. We define the relation $\sim$ on C_X as follows

$$x_n \sim y_n \iff \lim_{n\to\infty} d(x_n, y_n) = 0$$

Not that the RHS exists by the above. Clearly $\sim$ is an equivalence relation. Let $\hat{X} = C_X / \sim$ (the set of equivalence classes), i.e.

$$\hat{X} = \{[x_n] \colon x_n \in C_X\}$$

We define $\hat{d}([x_n], [y_n]) = \lim_{n\to\infty} d(x_n, y_n)$ (this exists by the corollary above). Note that this is well-defined, indeed let $x_n \sim x'_n$ and $y_n \sim y'_n$ so we have that

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \to 0$$

Therefore

$$\lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(x'_n, y'_n)$$

We now show that $\hat{d}$ is a metric (clear since d is a metric):

- Clearly $\hat{d}([x_n],[y_n]) \geq 0$ and equal to zero iff $\lim_{n\to\infty} d(x_n,y_n) = 0 \iff x_n \sim y_n \iff [x_n] = [y_n]$.
- Symmetry follows from Symmetry of d.
- Triangle inequality follows from triangle inequality on d.

Define $\varphi\colon X \to \hat{X}$ as follows,

$$\varphi(x) = [(x,x,x,\cdots)], \quad (x,x,x,\cdots) \in C_x$$

This is clearly an isometry, we denote by $M := \varphi(X) = \{[(x,x,x,\cdots)\colon x \in X]\}$. We now show that $\overline{M} = \hat{X}$. Consider an element $[x_n] = [(x_1,x_2,\cdots)] \in \hat{X}$. Let $\varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\forall m,n \geq N$

$$d(x_n,x_m) < \varepsilon/2$$

Consider the sequence $P_n = (x_N,x_N,\cdots)$, clearly

$$d(x_n,P_n) = d(x_n,x_N) < \varepsilon/2, \quad \forall n \geq N$$

Then

$$\hat{d}([x_n],[P_n]) = \lim_{n\to\infty} d(x_n,P_n) \leq \varepsilon/2 < \varepsilon$$

With

$$[P_n] = [(x_N,x_N,\cdots)] \in M$$

Therefore $\overline{M} = \hat{X}$. It remains to show that $\hat{X}$ is complete, we use the lemma. Take a Cauchy sequence in M,

$$P_{x_1} = [(x_1,x_1,\cdots)],\ P_{x_2} = [(x_2,x_2,\cdots)],\ \cdots,\ P_{x_k} = [(x_k,x_k,x_k,\cdots)]$$

Construct the sequence $p = (x_1,x_2,x_3,\cdots)$, note that

$$d(x_n,x_m) = \hat{d}([(x_n)],[(x_m)]) = \hat{d}(P_{x_n},P_{x_m})$$

But P_{x_n} is Cauchy in $\hat{X}$, hence p is Cauchy in X, so $P := [p] \in \hat{X}$. Take $\varepsilon > 0$, then $\exists N \in \mathbb{N}$ s.t.

$$d(x_n,x_k) \leq \varepsilon/2, \quad \forall n,k \geq N$$

So we get that for $k \geq N$

$$\begin{aligned}\hat{d}(P,P_{x_k}) &= \lim_{n\to\infty} d(x_n,x_k)\\ &\leq \varepsilon/2\\ &< \varepsilon\end{aligned}$$

Hence

$$\lim_{k\to\infty} P_{x_k} = P \in \hat{X}$$

□

1.2 Compactness

Definition 1.38. (X, τ) topological spaces and $K \subseteq X$. Then K is compact iff every open cover $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ of K has a finite subcover.

Example 1.39. $[0, 1)$ is not compact in $(\mathbb{R}, |\cdot|)$, $\mathcal{G} = \{(-1, 1 - 1/n) \colon n \in \mathbb{N}\}$ has no finite subcover.
Finite sets are clearly compact.

Proposition 1.40. *(X, τ) topological space, then every closed subset of a compact set is compact.*

Proof. Let K be compact, and $E \subseteq K$ is closed. Let $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ open cover of E, so

$$E \subseteq \bigcup_{\alpha \in I} G_\alpha$$

But E^c is open with

$$K \subseteq \left(\bigcup_{\alpha \in I} G_\alpha \right) \cup E^c$$

Hence $\{G_\alpha\} \cup \{E^c\}$ is an open cover of K, so it has a finite subcover

$$K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_N} \cup E^c$$

Then clearly

$$E = E \cap K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_N}$$

so E is compact. □

Proposition 1.41. *Let (X, τ) be a Hausdorff space, then K compact $\implies$ K is closed.*

Proof. Take $q \in K^c$, for every $p \in K$ by Hausdorff $\exists U_p, V_p^q$ open neighborhoods of p, q resp. s.t. $U_p \cap V_p^q = \varnothing$. Take $\mathcal{G} = \{U_p\}_{p \in K}$ an open cover of K, so it has a finite subcover

$$K \subseteq U_{p_1} \cup \cdots \cup U_{p_N}$$

Now define

$$V^q = V_{p_1}^q \cap \cdots \cap V_{p_N}^q$$

open set with

$$V^q \cap K = \varnothing$$

So K is closed. □

Proposition 1.42. *(X, d) metric space, $K \subseteq X$ is compact $\implies$ K is closed and bounded.*

Proof. Ex. □

Theorem 1.43 (Heine Borel)**.** *The converse is true for $(\mathbb{R}^n, |\cdot|)$ or more generally f.d. vector spaces.*

Example 1.44. $\ell^\infty(\mathbb{R})$ $d_\infty(x,y) = \sup_{x_i - y_i}$ and $A = \{0,1\}^{\mathbb{N}}$. Note that A is bounded, since if $x_0 = (0,0,0,0,\cdots)$, notice that $d_\infty(x,x_0) \leq 1$, so A is bounded. Moreover, every convergent sequence in A is eventually a constant and so belongs to A so A is closed. We could also say that A^c is open, take $x \notin A$, then $\exists x_i \neq 0,\ x_i \neq 1$. So $\exists \varepsilon > 0$ s.t. forall $y \in (x_i - \varepsilon, x_i + \varepsilon)$ $y_0 \neq 0,\ y \neq 1$. take $z \in B(x,\varepsilon)$ so

$$|z_i - x_i| \leq d_\infty(z,x) < \varepsilon$$

i.e. $z_i \neq 0$ and $z_i \neq 1$ so $z \in A^c$.

Now A is not compact. Take $\mathcal{G} = \{B(x,1/2)\}_{x\in A}$, now $B(x,1/2) \cap A = \{x\}$ but A is infinite (uncountable even), then $\mathcal{G}$ does not have a finite subcover.

Definition 1.45. (X,d) metric space and $K \subseteq X$, then K is totally bounded iff $\forall \varepsilon > 0,\ \exists x_1, x_2, \cdots, x_n \in K$ s.t. $K \subseteq B(x_1,\varepsilon) \cup B(x_2,\varepsilon) \cup \cdots \cup B(x_n,\varepsilon)$.

Example 1.46. $\{0,1\}^{\mathbb{N}}$ is not totally bounded in $\ell^\infty(\mathbb{R})$ (although it is bounded).

Proposition 1.47. *K is totally bounded $\implies$ K is bounded.*

Proof. easy. □

Proposition 1.48. *A subset of a totally bounded set is totally bounded.*

Proof. Let K be totally bounded and $E \subseteq K$, fix $\varepsilon > 0$ then we have $x_1, x_2, \cdots, x_n \in K$ s.t.

$$K \subseteq B(x_1,\varepsilon/2) \cup \cdots \cup B(x_n,\varepsilon/2)$$

Note that $E \subseteq K$, so pick $y_1, \cdots, y_k \in B(x_i,\varepsilon/2)$ (in case of empty just skip). Then we have that

$$E \subseteq B(x_1,\varepsilon) \cup \cdots \cup B(x_k,\varepsilon)$$

□

Lemma 1.49. *Assume that x_n is a Cauchy sequence in (X,d) with a subsequence that converges to x_0, then $x_n \to x_0$.*

Proof. easy. □

Remark 1.50. In $\mathbb{R}^n$ (finite dimensional vector spaces), K is totally bounded iff K is bounded.

Proof. Let K be bounded, then $\exists Q$ a closed cube in $\mathbb{R}^n$ s.t. $K \subseteq Q$. Now Q is compact (closed and bounded), so given $\varepsilon > 0$ it can be covered by finitely many balls of radius ε and center in $\mathbb{Q}$. So the cube is totally bounded, and hence K is a subset of a totally bounded set, hence is totally bounded. □

Theorem 1.51. *(X,d) metric space and $K \subseteq Q$, then the following are equivalent:*

i) K is compact

ii) Every infinite subset E of K has a limit point in K.

iii) Every sequence in K has a convergent subsequence (with its limit in K).

iv) K is complete and totally bounded.

(Proof.) $(i) \implies (ii)$ Let $E \subset K$ be infinite s.t. E has no limit points in K. Then for each $x \in K$, $x \notin E'$ so $\exists \varepsilon_x$ s.t.

$$B^*(x, \varepsilon_x) \cap E = \varnothing$$

Now take $\mathcal{G} = \{B(x, \varepsilon_x)\}_{x \in K}$, this is an open cover of K so it has a finite subcover. We get

$$K \subseteq B(x_1, \varepsilon_{x_1}) \cup \cdots \cup B(x_N, \varepsilon_{x_N})$$

But $E \subseteq K$ so

$$E = E \cap K \subseteq \bigcup_{i=1}^{N} E \cap B(x_1, \varepsilon_{x_i}) \subseteq \{x_1, \cdots, x_N\}$$

But E is infinite, contradiction.

$(ii) \implies (iii)$ Let x_n be a sequence in K and define

$$E = \{x_n \colon n \in \mathbb{N}\}$$

If E is finite, by pigeonhole principle $\exists n_1 < n_2 < \cdots$ and $x_0 \in E$ s.t. $x_{n_k} = x_0 \in E$ so we have found a convergent subsequence. If not then by the hypothesis E has a limit point $x_0 \in K$. We know that

$$B(x_0, 1) \cap E \neq \varnothing$$

So let n_1 be such that $x_{n_1} \in B(x_0, 1) \cap E$. In fact if $x \in E'$ we have that $B(x, \varepsilon) \cap E$ is infinite (since we are in a metric space). Hence for each k,

$$B^*(x_0, 1/k) \cap E \text{ is infinite}$$

So we can find some $n_k > n_{k-1}$ with $d(x_{n_k}, x_0) < 1/k$, we have that

$$\lim_{k \to \infty} x_{n_k} = x_0$$

$(iii) \implies (iv)$ Let x_n be a Cauchy sequence in K. But K is sequentially compact, then x_n has a convergent subsequence $x_{n_k} \to x_0 \in K$. Now by the lemma, x_n converges to x_0 and so K is complete. Assume K is not totally bounded, then $\exists \varepsilon_0 > 0$ s.t. K cannot be covered by finitely many ε_0-balls with center in K. take $x_1 \in K$, then take $x_2 \in K \setminus B(x_1, \varepsilon_0)$, then $x_3 \in K \setminus (B(x_1, \varepsilon_0) \cup B(x_2, \varepsilon_0))$. Similarly take

$$x_k \in K \setminus (B(x_1, \varepsilon_0) \cup \cdots \cup B(x_{k-1}, \varepsilon_0))$$

Consider the sequence $\{x_n\}$, then for $n \neq m$ we have

$$d(x_n, x_m) \geq \varepsilon_0$$

So x_n has no convergent subsequence, contradiction.

$(iv) \implies (i)$ Assume K is not compact, then $\exists \mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ an open cover of K that does not have a finite subcover. Take $\varepsilon = 1$, so K can be covered by finitely many 1-balls. So a$\exists x_1 \in K$ and $B(x_1, 1)$ s.t. $K_1 := B(x_1, 1) \cap K$ cannot be covered by a finite subcover of $\mathcal{G}$. Now take $\varepsilon = 1/2$, since $K_1 \subseteq K$ is totally bounded we can similarly find $x_2 \in K_1$ with $K_2 := B(x_2, 1/2) \cap K_1$ cannot be covered by finitely many G_α's. Recursively, we construct

$$K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$$

with $K_n := B(x_n, 1/n) \cap K_{n-1}$ with $x_n \in K_{n-1}$ and K_n cannot be covered by finitely many elements in $\mathcal{G}$. We show that x_n is Cauchy. Fix $\varepsilon > 0$, take N s.t. $2/N < \varepsilon$, then for $n, m > N$ then

$$x_n \in K_{n-1} \subseteq K_N, \quad x_m \in K_{m-1} \subseteq K_N \implies x_n, x_m \in K_N = B(x_n, 1/N) \cap K_{N-1}$$

So

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_N) + d(x_N, x_m) \\ &< 2/N \\ &< \varepsilon \end{aligned}$$

So x_n is Cauchy.

Therefore by completeness, x_n converges to some $x_0 \in K$. But we must have that $\exists \alpha_0 \in I$ s.t. $x_0 \in G_{\alpha_0}$ which is open, so $\exists \delta_0 > 0$ s.t. $B(x_0, \delta_0) \subseteq G_{\alpha_0}$. Since $x_n \to x_0$, then $\exists N$ s.t. $1/N < \delta_0/2$ and $d(x_n, x_0) < \delta_0/2$. So take $y \in B(x_n, 1/N)$ so

$$\begin{aligned} d(y, x_0) &\leq d(y, x_n) + d(x_n, x_0) \\ &< 1/N + d(x_n, x_0) \\ &< \delta_0 \end{aligned}$$

So

$$B(x_n, 1/N) \subseteq B(x_0, \delta_0)$$

We get that $K_N = B(x_N, 1/N) \cap K_{N-1} \subseteq B(x_0, \delta_0) \subseteq G_{\alpha_0}$. □

Proposition 1.52. *$f\colon (X, \tau_X) \to (Y, \tau_Y)$ and $K \subseteq X$ compact, then $f(K)$ is compact.*

Proof. easy. □

1.3 The Arzela-Ascoli theorem

Definition 1.53. Let X be a compact metric space, we denote by $C(X)$ the space of real-valued continuous functions on X. We define

$$d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)|$$

Proposition 1.54. *$C(X)$ is complete and separable.*

Proof. Uniformly Cauchy in $\mathbb{R} \implies$ uniformly continuous, follows from the general Stone-Weierstrass theorem. □

Remark 1.55. Since X is compact, then every continuous function is uniformly continuous. I.e. $\forall \varepsilon > 0,\ \exists \delta > 0$ s.t.

$$d_Y(f(x), f(x')) < \varepsilon, \quad \forall d(x, x') < \delta$$

Definition 1.56. $\mathcal{F} \subseteq C(X)$ is said to be equicontinuous iff $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.

$$d(f(x), f(x')) < \varepsilon, \quad \forall x, x'\ d(x, x') < \delta,\ \forall f \in \mathcal{F}$$

Theorem 1.57 (Arzela-Ascoli)**.** *$\mathcal{F} \subseteq C(X)$ is totally bounded $\iff$ $\mathcal{F}$ is bounded and equicontinuous.*

Proof. $\implies$ $\mathcal{F}$ is bounded since it is totally bounded. Let $\varepsilon > 0$, since $\mathcal{F}$ is totally bounded, there exist $f_1, f_2, \cdots, f_N \in \mathcal{F}$ s.t.

$$\mathcal{F} \subseteq B(f_1, \varepsilon/3) \cup \cdots \cup B(f_N, \varepsilon/3)$$

Now $f_i \in C(X)$, so uniformly continuous, and so $\exists \delta_i > 0$ s.t.

$$|f_i(x) - f_i(y)| < \varepsilon/3, \quad \forall d(x, y) < \delta_i$$

Let $\delta = \min_i \delta_i$. Now take $f \in \mathcal{F}$, and $x, y \in X$ with $d(x, y) < \delta$, now $\exists i_0 \in \{1, \cdots, N\}$ s.t. $f \in B(f_{i_0}, \varepsilon/3)$. Hence

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_{i_0}(x)| + |f_{i_0}(x) - f_{i_0}(y)| + |f_{i_0}(y) - f(y)| \\ &\leq 2 \times d_\infty(f, f_{i_0}) + \varepsilon/3 \\ &< \varepsilon \end{aligned}$$

Therefore $\mathcal{F}$ is equicontinuous.

$\impliedby$ Let $\varepsilon > 0$, then $\exists \delta > 0$ s.t $\forall f \in \mathcal{F}$ we have

$$|f(x) - f(y)| < \varepsilon/3, \quad \forall x, y \in X \text{ s.t. } d(x, y) < \delta$$

Now X is compact, then we can cover X with finitely many δ-balls, i.e.

$$X = B(x_1, \delta) \cup \cdots \cup B(x_N, \delta)$$

Define a map $\varphi \colon \mathcal{F} \to \mathbb{R}^N$,

$$\varphi(f) = (f(x_1), \cdots, f(x_N))$$

We want to show that $\varphi(\mathcal{F})$ is bounded in $\mathbb{R}^N$. We know that $\mathcal{F}$ is bounded in $C(X)$, so $\exists M > 0$ s.t.

$$|f(x)| \leq M, \quad \forall f \in \mathcal{F},\ \forall x \in X$$

Therefore

$$|\varphi(f)| = \sqrt{f(x_1)^2 + \cdots + f(x_N)^2} \leq \sqrt{N}M$$

Therefore $\varphi(\mathcal{F})$ is bounded in $\mathbb{R}^N$. Therefore $\varphi(\mathcal{F})$ is totally bounded. Then there exists $f_1, f_2, \cdots, f_K \in \mathcal{F}$ s.t.

$$\varphi(\mathcal{F}) \subseteq B(\varphi(f_1), \varepsilon/3) \cup \cdots \cup B(\varphi(f_K), \varepsilon/3)$$

We want to show that

$$\mathcal{F} \subseteq B(f_1, \varepsilon) \cup \cdots \cup B(f_K, \varepsilon)$$

Let $f \in \mathcal{F}$, then $\varphi(f) \in \varphi(\mathcal{F})$ i.e. $\exists i_0$ s.t. $\varphi(f) \in B(\varphi(f_{i_0}), \varepsilon/3)$. Recall that

$$\varphi(f) = (f(x_1), \cdots, f(x_N))$$

But notice that for each j, we have

$$|f(x_j) - f_{i_0}(x_j)| < \varepsilon/3, \quad \forall j = 1, \cdots, N$$

Let $x \in X$, then $x \in B(x_{j_0}, \delta)$ for some

$$\begin{aligned} |f(x) - f_{i_0}(x)| &\leq |f(x) - f(x_{j_0})| + |f(x_{j_0}) - f_{i_0}(x_{j_0})| + |f_{i_0}(x_{j_0}) + f_{i_0}(x)| \\ &< \varepsilon \end{aligned}$$

Since this holds for each $x \in X$ we have

$$|f(x) - f_{i_0}(x)| < \varepsilon, \quad \forall x \in X$$

Then since X is compact

$$\sup|f(x) - f_{i_0}(x)| = \max|f(x) - f_{i_0}(x)| < \varepsilon$$

Hence $f \in B(f_{i_0}, \varepsilon)$, therefore $\mathcal{F}$ is totally bounded.

□

Corollary 1.58. *$\mathcal{F} \subseteq C(X)$ is compact iff $\mathcal{F}$ is closed and bounded and equicontinuous.*

Lemma 1.59. *$\mathcal{F} \subseteq C(X)$ is equicontinuous, then $\overline{\mathcal{F}}$ is equicontinuous.*

Proof. Take $\varepsilon > 0$ and $f \in \overline{f}$, then $\exists f_n \in \mathcal{F}$ s.t.

$$\sup_X |f_n - f|$$

but $\mathcal{F}$ is equicontinuous, so $\exists \delta > 0$ s.t. $\forall h \in \mathcal{F}$

$$|h(x) - h(x')| < \varepsilon/3, \quad \forall x, x' \text{ s.t. } d(x, x') < \delta$$

for $d(x, x') < \delta$

$$\begin{aligned} |f(x) - f(x')| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(x')| + |f_n(x') - f(x')| \\ &\leq 2 \sup_X |f - f_n| + \varepsilon/3 \end{aligned}$$

But $\exists N$ s.t. for $n \geq N$ $\sup|f - f_n| < \varepsilon/3$ and

$$|f(x) - f(x')| < \varepsilon$$

□

Theorem 1.60 (Sequential Arzela-Ascoli)**.** *Let f_n be a sequence in $C(X)$ s.t.*

i) f_n is uniformly bounded, i.e. $|f_n(x)| \leq M,\ \forall x \in X,\ \forall n \in \mathbb{N}$.

ii) f_n is equicontinuous, i.e. $\forall \varepsilon > 0,\ \exists \delta$ s.t.

$$|f_n(x) - f_n(x')| < \varepsilon,\ \forall x, x'\ d(x, x') < \delta,\ \forall n \in \mathbb{N}$$

Then f_n has a subsequence that converges uniformly to a function $f \in C(X)$.

Proof. $\mathcal{F} = \{f_n \in \mathbb{N}\}$, then $\overline{\mathcal{F}}$ is closed, bounded and equicontinuous. Then $\overline{\mathcal{F}}$ is compact and hence is sequentially compact. □

Example 1.61. In $C([0, 1])$, let $\mathcal{F} = \{f \in \mathrm{Lip}_1([0, 1]) \colon f(0) = \alpha\}$ where

$$\mathrm{Lip}_1([0, 1]) = \{f \colon |f(x) - f(y)| \leq |x - y|,\ \forall x, y \in X\}$$

Now $\mathcal{F}$ is compact in $C(X)$.

- **Closed.** Take $f_n \in \mathcal{F}$ s.t. $f_n \to f$ in $C([0, 1])$. Clearly $f_n(0) = \alpha$ hence $f(0) = \alpha$. Now

  $$|f_n(x) - f_n(y)| \leq |x - y|$$

  let $n \to \infty$ we have

  $$|f(x) - f(y)| \leq |x - y|$$

  Hence $f \in \mathcal{F}$.
- **Bounded.** Now let $f \in \mathcal{F}$, then $|f(x) - f(0)| \leq |x - \alpha| \leq 1 + \alpha$, therefore for any $f \in \mathcal{F}$ and $x \in X$ we have

  $$|f(x)| \leq |f(x) - f(0)| + |f(0)| \leq 1 + 2\alpha$$

  Therefore

  $$\sup|f(x)| \leq 1 + 2\alpha$$

  So $\mathcal{F}$ is bounded.
- **Equicontinuous.** Let $\varepsilon > 0$, take $\delta = \varepsilon$, then for $|x - y| < \delta = \varepsilon$ we have

  $$|f(x) - f(y)| \leq |x - y| < \varepsilon$$

By Arzela-Ascoli, we get that $\mathcal{F}$ is compact.

Example 1.62. Now fix $f \in C([0,1])$, and look at $m = \inf_g \{d_\infty(f,g) \in \mathrm{Lip}_1([0,1]) \colon g(0) = f(0)\}$, want to show that this is attained. Indeed, let g_n be a sequence in $\mathrm{Lip}_1([0,1])$ $g_n(0) = f(0)$ with $d_\infty(f, g_n) \to m$. By compactness, g_n has convergent subsequence g_{n_k} in $C(X)$ to a function $g \in \mathrm{Lip}_1([0,1])$. Now $d_\infty(f,g) = \lim_{k\to\infty} d_\infty(f, g_{n_k}) = m$.

1.4 Baire Category Theorem

Definition 1.63. (X,τ) topological space, $A \subseteq X$. We say that A is nowhere dense in X iff $\overline{A}$ has empty interior.

Example 1.64. In $\mathbb{R}$, all singletons (or discrete sets) are nowhere dense in $\mathbb{R}$. But $\mathbb{Q}$ is not nowhere dense in $\mathbb{R}$.

Proposition 1.65. *A is nowhere dense in $X \iff X \setminus \overline{A}$ is dense.*

Proof. $\implies$ take $x \in X$ and $U_x \ni x$ open neighborhood, but x is not an interior point of $\overline{A}$ because $\mathring{\overline{A}} = \varnothing$, therefore $U_x \cap \overline{A} \neq \varnothing$.

$\impliedby$ Same.

□

Remark 1.66. A closed set is nowhere dense iff $\mathring{A} = \varnothing$.

Definition 1.67. (X,τ) topological space and $A \subseteq X$. We say that A is first Category or meagre in X iff A can be written as a countable union of nowhere dense subsets of X. Otherwise we say that A is second category (or non meagre).

Example 1.68. $\mathbb{Q}$ is first category in $\mathbb{R}$.

Remark 1.69. If a set A is second Category with $A = \bigcup_{n=1}^\infty F_n$ with F_n closed, then $\exists n_0$ with F_{n_0} has nonempty interior.

Theorem 1.70 (Baire Category Theorem)**.** *(X,d) complete metric space, then if $G_1, G_2, \cdots$ are open dense subsets of X then $\bigcap_{n=1}^\infty G_n$ is dense in X. Equivalently, every nonempty open subset of X is second Category. In particular, if $X = \bigcup_{n=1}^\infty F_n$ with F_n closed, then $\exists n_0$ with F_{n_0} has nonempty interior.*

Why equivalently?

$\implies$ Let G open in X, suppose that we can write $G = \bigcup_{n=1}^\infty A_n$ where A_n is nowhere dense. Take $G_i = X \setminus \overline{A}$, note that G_i is open and dense in X, therefore $\bigcap_{n=1}^\infty G_n$ is dense in X. But

$$\begin{aligned}\bigcap_{n=1}^{\infty} G_n &= \bigcap_{n=1}^{\infty} (\overline{A_n})^c \\ &= \left(\bigcup_{n=1}^{\infty} \overline{A_n}\right)^c\end{aligned}$$

Which is dense in X. But we have that

$$\bigcup_{n=1}^{\infty} \overline{A_n} \supseteq G$$

with G open, contradiction.

$\Longleftarrow$ Let $G_1, G_2, \cdots$ be open dense subsets of X. Note that G_i is second Category, and suppose $\bigcap_{n=1}^{\infty} G_n$ is not dense, so $\exists H \subseteq (\bigcap_{n=1}^{\infty} G_n)^c = \bigcup_{n=1}^{\infty} G_n^c$ with H open. But this means that

$$H = \bigcup_{n=1}^{\infty} G_n^c \cap H$$

with H second category, then $\exists n_0 \in \mathbb{N}$ and an open set $U \subseteq G_{n_0}^c \cap H \subseteq G_{n_0}^c$, contradiction.

Proof of Baire Category Theorem. Take $x_0 \in X$ and $\delta > 0$, take $B_0 = B(x_0, \delta)$. Now B_0 intersects G_1, and $\exists x_1 \in B_0 \cap G_1$ and $\delta_1 < \delta/2$ s.t.

$$B_1 = B(x_1, \delta_1) \subseteq \overline{B(x_1, \delta_1)} \subseteq B_0 \cap G_1$$

Now B_1 intersects G_2, so $\exists x_2 \in B_1 \cap G_2$ and $\delta_2 < \delta/2^2$ with

$$B_2 = B(x_2, \delta_2) \subseteq \overline{B(x_2, \delta_2)} \subseteq B_1 \cap G_2 \subseteq B_0 \cap G_1 \cap G_2$$

Continue this recursively to get $x_n \in B_{n-1} \cap G_n$ and $\delta_n < \delta/2^n$ s.t.

$$B_n = B(x_n, \delta_n) \subseteq \overline{B(x_n, \delta_n)} \subseteq B_0 \cap G_1 \cap \cdots \cap G_n$$

We show that x_n is a Cauchy sequence in X. Indeed, let $n, m > N$ for some $N \in \mathbb{N}$ then $x_n, x_m \in B_N = B(x_N, \delta_N)$, so

$$d(x_n, x_m) < 2\delta/2^N = \delta/2^{N-1}$$

Now let $\varepsilon > 0$, choose N s.t. $\delta/2^{N-1} < \varepsilon$, then

$$d(x_n, x_m) < \varepsilon$$

so x_n is Cauchy. By completeness, x_n converges to some $z \in X$. We show that $z \in (\bigcap_{n=1}^{\infty} G_n) \cap B_0$. Fix m, we have

$$x_n \in \overline{B_m}, \quad \forall n > m$$

Therefore $z \in \overline{B_m} \subseteq B_0 \cap (\bigcap_{k=1}^{m} G_k)$, since this holds for every m, we have

$$z \in B_0 \cap \left(\bigcap_{k=1}^{\infty} G_k \right)$$

□

2 Normed Spaces

Definition 2.1. Given a vector space V over $\mathbb{F}$, a norm is a map

$$\|\cdot\| : V \to \mathbb{R}$$

satisfying

i) $\|x\| \geq 0$ with equality iff $x = 0$.

ii) $\|\alpha x\| = |\alpha| \cdot \|x\|$, $\alpha \in \mathbb{F},\ x \in V$.

iii) $\|x + y\| \leq \|x\| + \|y\|,\ \forall x, y \in V$.

Example 2.2. $\ell^\infty(\mathbb{R})$ (set of bounded sequences) under coordinate-wise addition and scalar multiplication.

$$\|x\|_\infty = \sup_{i \in \mathbb{N}} |x_i| = d_\infty(x, 0)$$

$C(X)$ with X compact

$$\|f\|_\infty = \sup_{x \in X} |f(x)|$$

Proposition 2.3. *Every norm induces a metric*

$$d(x, y) = \|x - y\|$$

This allows us to deal with normed space as metric spaces w.r.t. induced metric.

Definition 2.4. V is a vector space and $\|\cdot\|_a,\ \|\cdot\|_b$ norms on V, we say that $\|\cdot\|_a,\ \|\cdot\|_b$ are equivalent iff they induce the same topology.

Proposition 2.5. *V vector space, $\|\cdot\|_a,\ \|\cdot\|_b$ norms on V. The following are equivalent:*

- *$\|\cdot\|_a,\ \|\cdot\|_b$ are equivalent.*
- *$\exists c_1, c_2 > 0$ s.t.*
$$c_2\|x\|_b \leq \|x\|_a \leq c_1\|x\|_b, \quad \forall x \in V$$
- *$x_n \to x_0$ in $(V, \|\cdot\|_a) \iff x_n \to x_0$ in $(V, \|\cdot\|_b)$.*

(*Proof.* $\Rightarrow$ (ii) $B_{\|\cdot\|_a}(0,1) = \{x \in V,\ \|x\|_a < 1\}$ is open in $(V, \|\cdot\|_a)$ then $B_{\|\cdot\|_a}(0,1)$ is open in $(V, \|\cdot\|_b)$. Therefore 0 is an interior point of $B_{\|\cdot\|_a}(0,1)$ w.r.t. the $\|\cdot\|_b$ norm. Therefore $\exists \delta > 0$ s.t.

$$B_{\|\cdot\|_b}(0, \delta) \subseteq B_{\|\cdot\|_a}(0, 1)$$

Now take $x \in V,\ x \neq 0$, then

$$\frac{\delta}{2} \frac{x}{\|x\|_b} \in B_{\|\cdot\|_b}(0, \delta) \subseteq B_{\|\cdot\|_a}(0, 1)$$

Hence

$$|\frac{\delta}{2} \frac{x}{\|x\|_b}|_a < 1 \implies \|x\|_b < \frac{2}{\delta}|x|_b$$

do the same by switching $\|\cdot\|_a$ and $\|\cdot\|_b$.

$(ii) \implies (iii)$ We have that

$$c_1\|x_n - x_0\|_b \leq \|x_n - x_0\|_a \leq c_1\|x_n - x_0\|$$

$(iii) \implies (i)$ Closed in $(V, \|\cdot\|_a) \iff$ Closed in $(V, \|\cdot\|_b)$ by sequential definition.

□

Definition 2.6. Take V finite dimensional vector space and $S = \{v_1, \cdots, v_n\}$ a basis of V. Take $x \in V$, then $\exists!\ r_1, r_2, \cdots, r_n$ s.t.

$$x = r_1v_1 + \cdots + r_nv_n$$

We define

$$\|x\|_\infty := \max_{1\leq i\leq n} |r_i|$$

We then have the following:

Proposition 2.7. *V is a f.d. vector space and $S = \{v_1, \cdots, v_n\}$ a basis:*

1. *$(V, \|\cdot\|_\infty)$ is complete.*
2. *$\{x \in V : |x|_\infty \leq M\}$ is compact.*
3. *$K \subseteq (V, \|\cdot\|_\infty)$ is compact iff K is closed and bounded.*

Proof. 1. Let x_k be a Cauchy sequence in V, then

$$x_k = r_{k1}v_1 + \cdots + r_{kn}v_n$$

Then

$$|r_{ki} - r_{li}| \leq \|x_k - x_l\|_\infty$$

Then the sequence $(r_{ki})_k$ is Cauchy in $\mathbb{R}$ for every i, then r_{ki} converges to some r_i. Take $x = r_1v_1 + \cdots + r_nv_n$, we get that

$$\begin{aligned}\|x_k - x\|_\infty &= \|(r_{k1} - r - 1)v_1 + \cdots + (r_{kn} - r_n)v_n\|_\infty \\ &\leq \|r_{k1} - r_1\|\|v_1\|_\infty + \cdots + \|r_{kn} - r_n\||v_n| \\ &\to 0\end{aligned}$$

2. Take a sequence x s.t. $\|x_k\|_\infty \leq M$. Apply Bolzano Weierstrass n times and take subsequence of the subsequence.

3. $(\implies)$ is always true. $(\impliedby)$ since K is bounded, then $\exists M$ s.t.

$$K \subseteq \{x \in V : \|x\|_\infty \leq M\} =: Q$$

But K is closed Q is compact, then K is compact.

□

Theorem 2.8. *V is a finite dimensional vector space, then all norms on V are equivalent.*

Corollary 2.9. *V finite dimensional and $\|\cdot\|$ is a norm on V then $(V, \|\cdot\|)$ has the properties of the theorem above.*

Moreover, if V is a normed space and M is a finite dimensional subspace of V.

Remark 2.10. Note that the converse is not true in general, take $P([a,b])$ as a subspace of $C([a,b])$.

Proof. Take $S = \{v_1, \cdots, v_n\}$ a basis of V, define

$$\|x\|_\infty = \max_{1 \le i \le N} |r_i|$$

Let $\|\cdot\|_a$ be a norm on V, we will show that $\|\cdot\|_a$ is equivalent to $\|\cdot\|$.

$$\begin{aligned} \|x\|_a &= \|r_1 v_1 + \cdots + r_n v_n\|_a \\ &\le |r_1| \|v_1\|_a + \cdots + |r_n| \|v_n\|_a \\ &\le (\|v_1\|_a + \cdots + \|v_n\|_a) \|x\|_\infty \end{aligned}$$

Let $\varphi \colon (V, \|\cdot\|_\infty) \to \mathbb{R}$, note that this is continuous since if $x_n \to x_0$ in $\|\cdot\|_\infty$, then

$$\begin{aligned} |\varphi(x_n) - \varphi(x_0)| &= (\|x_n\|_a - \|x_0\|) \\ &\le \|x_n - x_0\|_a \\ &\le c_1 \|x_n - x_0\|_\infty \\ &\to 0 \end{aligned}$$

Now let $K = \{x \in V \colon \|x\|_\infty = 1\}$, note that K is compact (closed and bounded), so $\varphi(K)$ is compact in $\mathbb{R}$ so $m = \inf \varphi(K)$ is attained at some $x_0 \in K,\ x_0 \neq 0$. Then $m = \|x_0\|_a > 0$. So $\forall x \in V$ s.t. $\|x\|_\infty = 1$ we have

$$|x|_a \ge m$$

Take $v \in V$ s.t. $x \neq 0$ we have that

$$\begin{aligned} \| \frac{x}{\|x\|_\infty} \|_a &\ge m \\ \|x\|_a &\ge m \|x\|_\infty \end{aligned}$$

□

Lemma 2.11 (Riesz Lemma)**.** *$(V, \|\cdot\|)$ is a normed v.s. and M a proper closed subspace of V. Then for every $0 < \alpha < 1$ there exists $x \in V$ s.t. $\|x\| = 1$ and*

$$d(x, M) \ge \alpha$$

Remark 2.12. If x is s.t. $\|x\| = 1$, then $d(x, M) \le |x - 0| \le 1$.

Proof. Take $x_0 \notin M$, $d(x_0, M) > 0$ since M is closed. Fix $0 < \alpha < 1$, there exists $m_0 \in M$ s.t.

$$d(x_0, M) \le \|x_0 - m_0\| < \frac{d(x_0, M)}{\alpha}$$

Define

$$x = \frac{x_0 - m_0}{\|x_0 - m_0\|}, \quad \|x\| = 1$$

Let $m \in M$, then

$$\begin{aligned}\|x - m\| &= \|\frac{x_0 - m_0}{\|x_0 - m_0\|} - m\| \\ &= \frac{\|x_0 - (m_0 + m\|x_0 - m_0\|)\|}{\|x_0 - m_0\|}\end{aligned}$$

But $m_0 + m\|x_0 - m_0\| \in M$, so

$$\|x - m\| \geq \frac{d(x_0, M)}{\|x_0 - m_0\|} > \alpha$$

Therefore

$$d(x, M) = \inf_{m \in M} \|x - m\| \geq \alpha$$

□

Theorem 2.13. *$(V, \|\cdot\|)$ normed space, the closed unit sphere in V is compact iff V is finite dimensional.*

Proof. $\Longleftarrow$ ok.

$\Longrightarrow$ Assume that V is infinite dimensional, then we can construct

$$E_1 = \text{Span}\{v_1\}, \ E_2 = \text{Span}\{v_1, v_2\}, \ \cdots$$

With E_i finite dimensional and $\{v_i\}$ linearly independent. Then

$$E_1 \subsetneq E_2 \subsetneq E_3 \subsetneq \cdots$$

Fix some $u_1 \in E_1$ with $\|u_1\| = 1$, now by Riesz' lemma $\exists u_2 \in E_2$ s.t. $\|u_2\| = 1$ and $d(u_2, E_1) \geq 1/2$. Now $\exists u_3 \in E_3$ with $\|u_3\| = 1$ and $d(u_3, E_2) \geq 1/2$, recursively we can construct $u_n \in E_n$ s.t. $\|u_n\| = 1$ and $d(u_n, E_{n-1}) \geq 1$. Now notice that for $m \neq n$ we have

$$\|u_m - u_n\| \geq 1/2$$

Therefore u_n is a sequence which belongs to the unit sphere that has no Cauchy (and hence convergent) subsequence. So the unit sphere is not sequentially compact, and hence not compact.

□

Definition 2.14. A complete normed space is called a Banach space.

Example 2.15. Some Banach spaces: $\mathbb{R}^N$. $\ell^\infty(\mathbb{R})$ (which is the set of bounded sequences). $\ell(\mathbb{R}) = \{x \colon (x_1, x_2, \cdots,) \colon \sum|x_i| < \infty\}$. $C(X)$ when X is compact.

$P(\mathbb{R})$ is not Banach.

ℓ^p Spaces.

Lemma 2.16. $1 < p, q < \infty$ *s.t.* $\frac{1}{p} + \frac{1}{q} = 1$ *and* $x, y > 0$ *s.t.* $xy = 1$ *then*

$$\frac{x^p}{p} + \frac{y^q}{q} \geq 1$$

Proof. $f(x, y) = \frac{x^p}{p} + \frac{y^q}{q}$, we want to minimize $f(x, y)$ on $g(x, y) = xy = 1$. Note that this minimum exists since this is a closed set and $f(x, y) \geq 0$ for all x, y s.t. $xy = 1$. Using Lagrange multipliers, we solve

$$\begin{cases} \nabla f &= \lambda \nabla g \\ xy &= 1 \end{cases}$$

i.e.

$$\begin{cases} x^{p-1} = \lambda y \\ y^{q-1} = \lambda y \\ xy = 1 \end{cases}$$

Then we get

$$x^p = \lambda xy = \lambda, \quad y^q = \lambda$$

Then $x = \lambda^{1/p}, y = \lambda^{1/q}$, then

$$\lambda^{\frac{1}{p}+1/q} = 1 \implies \lambda = 1$$

Then $x = y = 1$. So the minimum is $f(1, 1) = 1$. □

Corollary 2.17. $\frac{1}{p} + \frac{1}{q} = 1$, $x, y \geq 0$ *then*

$$\frac{x^p}{p} + \frac{y^q}{q} \geq xy$$

Proof. Take $X = \frac{x}{(xy)^{1/p}}$ and $Y = \frac{y}{(xy)^{1/q}}$ then

$$XY = \frac{xy}{(xy)^{1/p+1/q}} = 1$$

Then apply the lemma, we get

$$\begin{aligned} \frac{X^p}{p} + \frac{Y^q}{q} &\geq 1 \\ \frac{x^p}{(xy)p} + \frac{y^q}{(xy)q} &\geq 1 \\ \frac{x^p}{p} + \frac{y^q}{q} &\geq xy \end{aligned}$$

□

Theorem 2.18 (Young's inequality)**.** *Let* $x, y \in \mathbb{R}$ *then*

$$|xy| \leq \frac{|x|^p}{p} + \frac{|y|^q}{q}$$

with $\frac{1}{p}+\frac{1}{q}=1$.

Theorem 2.19 (Holder inequality)**.** *Let* $x_1, x_2, \cdots, x_n \in \mathbb{R}$ *and* $y_1, y_2, \cdots, y_n \in \mathbb{R}$ *then*

$$\sum_{i=1}^{n}|x_i y_i| \leq \left(\sum_{i=1}^{n}|x_i|^p\right)^{1/p}\left(\sum_{i=1}^{n}|y_i|^q\right)^{1/q}$$

With $\frac{1}{p}+\frac{1}{q}=1$.

Letting $n \to \infty$ *we have*

$$\sum_{n=1}^{\infty}|x_n y_n| \leq \left(\sum_{n=1}^{\infty}|x_n|^p\right)^{1/p}\left(\sum_{n=1}^{\infty}|y_n|^q\right)^{1/q}$$

Proof. Wlog suppose not all zeroes. Let $X_i = \frac{|x_i|}{\left(\sum_{k=1}^{n}|x_k|^p\right)^{1/p}}$ and $y_i = \frac{|y_i|}{\left(\sum_{k=1}^{n}|y_k|^q\right)^{1/q}}$. From Young's inequality, we have that

$$\begin{aligned} X_i Y_i &= \frac{|x_i||y_i|}{\left(\sum_{k=1}^{n}|x_k|^p\right)^{1/p}\left(\sum_{k=1}^{n}|y_k|^q\right)^{1/q}} \\ &\leq \frac{X_i^p}{p}+\frac{Y_i^q}{q} \\ &= \frac{|x_i|}{\left(\sum_{k=1}^{n}|x_k|^p\right)p}+\frac{|y_i|^q}{\left(\sum_{k=1}^{n}|y_k|^q\right)q} \end{aligned}$$

Therefore

$$\begin{aligned} \sum_{i=1}^{n} X_i Y_i &\leq \frac{1}{p}+\frac{1}{q} \\ &= 1 \end{aligned}$$

Substituting we get the desired inequality. □

Theorem 2.20 (Minkowski inequality)**.** $(|x_i+y_i|)^p \leq \left(\sum_{i=1}^{n}|x_i|^p\right)^{1/p}+\left(\sum_{i=1}^{n}|y_i|^p\right)^p$. *In the infinite case we get*

$$\left(\sum_{n=1}^{\infty}|x_n+y_n|^p\right)^{1/p} \leq \left(\sum_{n=1}^{\infty}|x_n|^p\right)^{1/p}+\left(\sum_{n=1}^{\infty}|y_n|^p\right)^{1/p}$$

Proof. If $p = 1$, trivial. Take $p > 1$ and let q be s.t. $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\begin{aligned}
\sum_{i=1}^{n} |x_i + y_i|^p &= \sum_{i=1}^{n} |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\
&\leq \sum_{i=1}^{n} \left(|x_i| + |y_i|\right) |x_i + y_i|^{p-1} \\
&= \sum_{i=1}^{n} |x_i||x_i + y_i|^{p-1} + \sum_{i=1}^{n} |y_i||x_i + y_i|^{p-1} \\
&\leq \left(\sum_{i=1}^{n} |x_i|\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q} \\
&\quad + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q}
\end{aligned}$$

We have that $\frac{1}{p} + \frac{1}{q} = 1 \implies p + q = pq \implies p = q(p-1)$, therefore

$$\sum_{i=1}^{n} |x_i + y_i|^p \leq \left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/q} \left(\left(\sum_{i=1}^{n} |x_i|^{[}\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}\right)$$

$$(\sum_{i=1}^{n} |x_i + y_i|^p)^{1-1/q=1/p} \leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}$$

as desired. □

Definition 2.21. Let $p \geq 1$, we define the space $\ell^p(\mathbb{R}) = \{x = (x_1, \cdots,) \colon \sum_{n=1}^{\infty} |x_n|^p < \infty\}$ with the following norm $\|x\|_{\ell^p} = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$.

Proposition 2.22.

- *$\ell^p(\mathbb{R})$ is complete $\forall 1 \leq p \leq \infty$.*
- *$\ell^p(\mathbb{R})$ is separable $\forall 1 \leq p < \infty$ (recall that $\ell^\infty(\mathbb{R})$ is not separable). Since $c_{00}(\mathbb{Q})$ is dense in $\ell^p(\mathbb{R})$.*
- *$K \subseteq \ell^p(\mathbb{R})$ ($p \leq 1 < \infty$) is compact iff it is closed and bounded and*

$$\lim_{N \to \infty} \sup_{x \in K} \sum_{N=1}^{\infty} |x_n|^p = 0$$

- *$1 \leq r \leq s < \infty$, then*

$$\ell^r(\mathbb{R}) \subseteq \ell^s(\mathbb{R})$$

Hence

$$\ell^1(\mathbb{R}) \subseteq \ell^r(\mathbb{R}) \subseteq \ell^s(\mathbb{R}) \subseteq \ell^\infty(\mathbb{R})$$

- *If $1 \leq r < s < \infty$ then $\ell^r(\mathbb{R}) \subsetneq \ell^s(\mathbb{R})$*

$$x_n = \frac{1}{n^{1/r}}$$

Then $\sum |x_n| < \infty$ but $\sum |x_n|^r = \infty$ so $x_n \in \ell^s(\mathbb{R})$ and $x_n \notin \ell^r(\mathbb{R})$.

2.1 Bounded Linear Transformation

Definition 2.23. V, W vector spaces and $\mathcal{L}(V, W)$ is the space of linear transformations $T\colon V \to W$, this is a vector space.

Remark 2.24. If V, W are finite dimensional then $\mathcal{L}(V, W)$ is finite dimensional with

$$\dim \mathcal{L}(V, W) = \dim(V) \cdot \dim(W)$$

Definition 2.25. Take $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ and $T \in \mathcal{L}(V, W)$ then we say that T is bounded iff $\exists M > 0$ s.t.

$$\|Tx\|_W \leq M\|x\|_V$$

Equivalently $\exists M$ s.t.

$$\|Tx\| \leq M, \quad \forall x \text{ s.t. } \|x\|_V = 1$$

Proposition 2.26. *$(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ normed spaces and $T \in \mathcal{L}(V, W)$, then*

$$T \text{ bounded} \iff T \text{ is continuous} \iff T \text{ is continuous at } 0$$

(*Proof.*) (ii) Then $\exists M$ s.t. $\|Tx\| \leq M\|x\|$. Then

$$\begin{aligned} \|Tx - Ty\| &= \|T(x - y)\| \\ &\leq M\|x - y\| \end{aligned}$$

Then T is Lipschitz so it is continuous.

$(ii) \implies (iii)$ ok.

$(iii) \implies (i)$ Let T be continuous at 0, then $\exists \delta > 0$ s.t. $\|Tx - T0\| < 1$ for $\|x - 0\| < \delta$ but this gives

$$\|Tx\| \leq 1$$

for $\|x\| < \delta$. Now take $x \neq 0$, then

$$\|T\left(\frac{\delta}{2}\frac{x}{\|x\|}\right)\| \leq 1$$

So we get

$$\|Tx\| \leq \frac{\delta}{2}\|x\|$$

□

Remark 2.27. If V is finite dimensional, and $T \in \mathcal{L}(V, W)$, then T is continuous. Take $x \in V$, and $S = \{v_1, \cdots, v_n\}$ basis of V. Then

$$x = r_1 v_1 + \cdots + r_n v_n \implies \|Tx\| \leq r_1\|Tv_1\| + \cdots + r_n\|Tv_n\| \leq (\|Tv_1\| + \cdots \|Tv_n\|)\|x\|_\infty$$

But all norms on V are equivalent, then

$$\|Tx\|_W \leq M\|x\|_V$$

Then T is bounded.

Definition 2.28. $(V, \|\cdot\|)$, $(W, \|\cdot\|)$ normed spaces, we define $B(V, W)$ to be the space of bounded linear transformations with the following norm

$$\|T\|_{B(V,W)} = \sup_{\substack{x\in V \\ \|x\|=1}} \|Tx\|_W = \sup_{\substack{x\in V \\ x\neq 0}} \frac{\|Tx\|}{\|x\|}$$

Check that this is subspace of $\ell(V, W)$ and $\|\cdot\|_{B(V,W)}$ is a norm.

Remark 2.29. Notice that for $x \in V$ $x \neq 0$ we have that

$$|T\frac{x}{\|x\|}| \leq \|T\|$$
$$\|Tx\| \leq \|T\| \cdot \|x\|$$

Proposition 2.30. *If W is Banach, then $B(V, W)$ is Banach.*

Proof. Let T_n be a Cauchy sequence in $B(V, W)$, note that

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \cdot \|x\|$$

But T_m is Cauchy in $B(V, W)$, therefore $T_n(x)$ is Cauchy is W (which is Banach), then in converges. So we define

$$T\colon V \to W, \quad T(x) = \lim_{n\to\infty} T_n(x)$$

We want to show T is linear, this is trivial by properties of limits and that T_n are linear. Moreover $\{T_n\}$ is Cauchy in $B(V, W)$, then it is bounded i.e. $\exists M > 0$ s.t. $\|T_n\| \leq M$, $\forall n$. Now

$$\|T_n x\| \leq \|T_n\| \cdot \|x\|$$
$$\|T_n x\| \leq M\|x\|$$

Letting $n \to \infty$ we get that $\|Tx\| \leq M\|x\|$ so $T \in B(V, W)$.

We want to show that $T_n \to T$ in $B(V, W)$. Given $\varepsilon > 0$, since T_n is Cauchy, so $\exists N_1$ s.t.

$$\|T_n - T_m\| < \varepsilon/2, \quad \forall n, m \geq N_1$$

Now fix $x \in V$, since $T_n x \to Tx$ in W we have that $\exists N_2(x) > N_1$ s.t.

$$\|T_{N_2} x - Tx\| < \varepsilon/2$$

Now for $n \geq N_1$ and $\|x\| = 1$ we have

$$\begin{aligned}\|T_n x - Tx\| &< \|T_n x - T_{N_2} x\| + \|T_{N_2} x - Tx\| \\ &\leq \|T_n - T_{N_2}\| \cdot \|x\| + \frac{\varepsilon}{2} \\ &< \varepsilon\end{aligned}$$

Therefore given $\varepsilon > 0$, $\exists N_1$ s..t $\forall x \in V$ with $\|x\| = 1$ then

$$\|T_n x - Tx\| < \varepsilon, \quad \forall n \geq N_1$$

Therefore

$$\sup_{\substack{x \in V \\ \|x\|=1}} \|T_n x - Tx\| \leq \varepsilon, \quad \forall n \geq N_1$$

Hence

$$T_n \to T \in B(V, W)$$

□

Example 2.31. Define

$$T\colon\ (C([0,1]), \|\cdot\|_\infty) \to \mathbb{R}, \quad T(f) = \int_0^1 f(x)\, dx$$

Show that $T \in B(C([0,1]), \mathbb{R})$ and find $\|T\|$.

Solution: Clearly T is linear because integration is linear. Now

$$\begin{aligned} |Tf| &= |\int_0^1 f(x)\, dx| \\ &\leq \int_0^1 |f(x)|\, dx \\ &\leq \|f\|_\infty \end{aligned}$$

Take $T \in B(C([0,1]), \mathbb{R})$, note that

$$\sup_{\|f\|_\infty} \|Tf\| \leq 1$$

moreover if we take $f = 1$ we have that $\|Tf\| = 1$. Therefore

$$\|T\| = 1$$

2.2 Duality

Definition 2.32. Let V normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then the dual set $V^* = B(V, \mathbb{F})$, which is the set of all bounded linear maps from $V \to \mathbb{F}$. In particular, the dual norm

$$\|f\|_* = \sup_{\substack{x \in V \\ \|x\|=1}} \|f(x)\| = \sup_{\substack{x \in V \\ x \neq 0}} \|\frac{|f(x)|}{\|x\|}\|$$

Remark 2.33. If V is finite dimensional, then $V^* = B(V, \mathbb{F}) = \mathcal{L}(V, \mathbb{F})$, then

$$\dim V^* = \dim V < \infty$$

Therefore V^* and V are isomorphic.

notation. Given $(V, \|\cdot\|)$, $(W, \|\cdot\|)$ we write $V \cong W$ iff V and W are isometrically isomorphic. i.e. $\exists \varphi \colon V \to W$ and isomorphism and isometry.

Example 2.34. $(\ell^1(\mathbb{R}))^* \cong \ell^\infty(\mathbb{R})$, more particularly if $1 < p < q < \infty$ and $1/p + 1/q = 1$, then

$$(\ell^p)^* \cong \ell^q$$

So for $p = 2$ we get that

$$(\ell^2)^* \cong \ell^2$$

Now if $1 < p < q < +\infty$ then

$$(\ell^p)^{**} \cong (\ell^q)^* \cong (\ell^p)^*$$

If $p = 1$, then

$$(\ell^1)^* \cong \ell^\infty \text{ but } (\ell^\infty)^* \neq \ell^1$$

Theorem 2.35. $(\ell^1)^* \cong \ell^\infty$, *moreover if* $1 < p < q < \infty$ *then*

$$(\ell^p)^* \cong \ell^q$$

Where $\frac{1}{p} + \frac{1}{q} = 1$

Lemma 2.36. *Let* $1 \leq p < \infty$ *and let* $x = (x_1, x_2, \cdots) \in \ell^p(\mathbb{R})$ *then*

$$x = \sum_{n=1}^{\infty} x_n e_n = \lim_{N \to \infty} \sum_{n=1}^{N} x_n e_n$$

Proof. Indeed note that

$$\sum_{n=1}^{N} x_n e_n = (x_1, x_2, \cdots, x_N, 0, 0, \cdots)$$

and since $x_n \in \ell^p(\mathbb{R})$ we have that

$$\sum_{n=1}^{\infty} |x_n|^p < \infty$$

Now

$$\begin{aligned} \|x - \sum_{n=1}^{N} x_n e_n\|_{\ell^p} &= \|(0, \cdots, 0, x_{N+1}, x_{N+2})\|_p \\ &= \left(\sum_{n=N+1}^{\infty} |x_n|^p \right)^{1/p} \\ &\to 0 \end{aligned}$$

□

Remark 2.37. Now for ℓ^∞ the result is false. Indeed consider

$$\mathbb{1} = (1, 1, 1, \cdots)$$

Then

$$\sum_{n=1}^{N} x_n e_n = (1, 1, \cdots, 1, 0, 0, \cdots)$$

But

$$\begin{aligned} \|x - \sum_{n=1}^{N} x_n e_n\|_\infty &= \|(0, \cdots, 0, 1, 1, \cdots)\|_\infty \\ &= 1 \end{aligned}$$

Which means that

$$\lim_{N\to\infty} \sum_{n=1}^{N} x_n e_n \neq x$$

Proof of $(\ell^1)^* \cong \ell^\infty$. Let $T \in (\ell^1)^*$, then $T\colon \ell^1 \to \mathbb{R}$ bounded. Let $x \in \ell^1$, then by the lemma we have that $x = \sum_{n=1}^{\infty} x_n e_n$. With T continuous and linear therefore

$$Tx = \sum_{n=1}^{\infty} x_n T e_n$$

Let $b = (Te_1, Te_2, \cdots)$, we want to show that $b \in \ell^\infty$. Indeed, since T is bounded we have

$$\|Te_i\| \leq \|T\| \cdot \|e_i\| = \|T\|$$

Therefore

$$\sup_{i\in\mathbb{N}} |Te_i| \leq \|T\|$$

Hence $b \in \ell^\infty$ with

$$\|b\|_\infty \leq \|T\|$$

On the other hand

$$\begin{aligned} |Tx| &= |x_n T e_n| \\ &\leq \sum_{n=1}^{\infty} |x_n| \cdot |b_n| \\ \|Tx\| &\leq \|b\|_\infty \|x\| \end{aligned}$$

Therefore

$$\|T\| \leq \|b\|_\infty$$

Therefore

$$\|T\|_* = \|b\|_\infty$$

Using all of this we know that the linear map

$$\psi\colon (\ell^1)^* \to \ell^\infty, \quad \psi(T) = (Te_1, Te_2, \cdots)$$

is an isometry. To prove surjectivity, we can take $b = (b_1, b_2, \cdots) \in \ell^\infty$ and define $T_b \colon \ell^1 \to \mathbb{R}$ by

$$T_b x = \sum_{n=1}^{\infty} x_n b_n$$

Note that this converges since

$$\sum_{n=1}^{\infty} |x_n b_n| \leq \|b\| \|x\|$$

Then T_b is linear and

$$|T_b x| \leq \|b\|_\infty \|x\| \implies T_b \in (\ell^1)^*$$

Then

$$\begin{aligned} \psi(T_b) &= (T_b e_1, T_b e_2, \cdots) \\ &= (b_1, b_2, \cdots) \\ &= b \end{aligned}$$

□

3 Hahn-Banach Theorems

Motivation. Let $(V, \|\cdot\|)$ be a normed space over $\mathbb{F}$ and let $\mathcal{M}$ be a subspace of V and $T \colon M \to \mathbb{F}$ a bounded linear map. Notice that

$$\|Tx - Ty\| \leq \|T\| \cdot \|x - y\|$$

Therefore T is uniformly continuous and $\mathbb{F}$ is complete, then by the homework the map

$$\tilde{T} \colon \overline{\mathcal{M}} \to \mathbb{F}, \quad \tilde{T}x = \lim_{n \to \infty} Tx_n, \ x_n \text{ a sequence in } \mathcal{M} \text{ converging to } x$$

is a continuous extension of T into $\overline{\mathcal{M}}$. Moreover, $\mathcal{M}$ is a subspace then $\overline{\mathcal{M}}$ is also a subspace. Then $\tilde{T}$ is also linear. Then $\tilde{T} \in B(\overline{\mathcal{M}}, \mathbb{F})$. Now by definition

$$\begin{aligned} \|\tilde{T}\| &= \sup_{\substack{x \in \overline{\mathcal{M}} \\ x \neq 0}} \frac{\|\tilde{T}x\|}{\|x\|} \\ &\geq \|T\| \end{aligned}$$

Let $x \in \overline{\mathcal{M}}$ and $x_n \to x$ with $x_n \in \mathcal{M}$. Then

$$|Tx_n| \leq \|T\| \cdot \|x_n\|$$

Letting $n \to \infty$ we get

$$\|\tilde{T}x\| \leq \|T\| \cdot \|x\| \implies \|\tilde{T}\| \leq \|T\|$$

Lemma 3.1. *$(V, \|\cdot\|)$ normed space and $\mathcal{M}$ proper subspace of V and $x_0 \notin \mathcal{M}$. Let*

$$f \colon \mathcal{M} \to \mathbb{R}$$

a bounded linear map and let $x_0 \notin \mathcal{M}$. Then there exists an extension

$$F \colon \mathcal{M} \oplus \operatorname{Span}\{x_0\} \to \mathbb{R}$$

s.t. F is a bounded linear map and

$$F_{|M} = f$$

and

$$\|F\| = \|f\|$$

Proof. Take $x \in \mathcal{M} \oplus \operatorname{Span}\{x_0\}$, then we can write

$$x = m + tx_0, \quad t \in \mathbb{R}$$

We would like to have

$$\begin{aligned} F(x) &= F(x + tx_0) \\ &= f(m) + tc \end{aligned}$$

So we need to define $F(x_0) := c \in \mathbb{R}$ s.t.

$$|F(m + tx_0)| \leq \|f\|\|m + tx_0\| (*)$$

Since this would give us that

$$\|F\| = \|f\|$$

Clearly we have that $(*)$ holds for $t = 0$. Note that it is enough to prove this for $t = 1$, i.e. that

$$|F(m + x_0)| \leq \|f\|\|m + x_0\|$$

Indeed, suppose that this is the case, then for $t \neq 0$

$$\begin{aligned} |F(m + tx_0)| &= |t||F(\frac{m}{t} + x_0)| \\ &\leq |t|\|f\|\|\frac{m}{t} + x_0\| \\ &= \|f\|\|m + tx_0\| \end{aligned}$$

Then it is enough to find c s.t.

$$|F(m + x_0)| \leq \|f\|\|m + x_0\|$$

Which is equivalent to saying that

$$-f(m) - \|f\|\|m + x_0\| \leq c \leq \|f\|\|m + x_0\| - f(m), \quad \forall m \in \mathcal{M}, \ (**)$$

Take $m, m' \in \mathcal{M}$ then

$$\begin{aligned} f(m') - f(m) &= f(m' - m) \\ &\leq |f(m' - m)| \\ &\leq \|f\|\|m' - m\| \\ &\leq \|f\| \left(\|m' + x_0\| + \|m + x_0\|\right) \end{aligned}$$

Which is equivalent to saying that

$$f(m) - \|f\|\|m + x_0\| \leq -f(m') + \|f\|\|m' + x_0\|, \quad \forall m, m' \in \mathcal{M}$$

Hence

$$\sup_m \left(-f(m) - \|f\|\|m + x_0\|\right) \leq \inf_m \left(-f(m) + \|f\|\|m + x_0\|\right)$$

And so we can find c satisfying $(**)$. Then we can define

$$F(m + tx_0) = f(m) + tc_0$$

Hence

$$|F(m + tx_0)| \leq \|f\|\|m + tx_0\|$$

Therefore F is bounded and clearly $\|F\| = \|f\|$ (this gives us $\leq$ but from the definition $\geq$ is clear). □

3.1 Partially Ordered Sets

Definition 3.2. Given a set X, a partial order on X is a relation $\preceq$ on X that is

i) Reflexive: $x \preceq x$.

ii) Antisymmetric: $x \preceq y$ and $y \preceq x \implies x = y$.

iii) Transitive: $x \preceq y$ and $y \preceq z \implies x \preceq z$.

Then $(X, \preceq)$ is a partially ordered set.

Example 3.3. We can say that $(\mathcal{P}(X), \subseteq)$ is a partially ordered set.

We can also define on $\mathbb{N}$ the relation

$$a \leq b \iff a \text{ divides b}$$

Notice that in this case $2 \leq 6$ but 2 and 3 are not comparable.

Definition 3.4. Given a partially ordered set $(X, \leq)$ we say that $(X, \leq)$ is totally ordered iff $\forall x, y \in X$ we have that $x \leq y$ or $y \leq x$.

Definition 3.5. Given a partially ordered set $(X, \leq)$, we say that $M \in X$ is a maximal element of X iff $\forall a \in X$ s.t. $a \geq M$ then $a = M$.

Let $A \subseteq X$ we say that $M \in X$ is an upper bound of A iff $\forall a \in A$ we have $a \leq M$.

Example 3.6. $\{2, 3, 5, 30\}$ with divisibility, then 30 is an upper bound for this set. But if we take $A = \{2, 3, 5, 6\}$ as a subset of $\mathbb{N}$, then 30 is an upper bound of A, but the maximal elements of A are $5, 6$.

Definition 3.7. Let $(X, \leq)$, be a partially ordered set. Then a chain C of X is totally ordered subset of X.

Theorem 3.8 (Zorn's Lemma)**.** *Let (X, τ) be a nonempty partially ordered set such that every chain in X has an upper bound. Then X has a maximal element.*

Theorem 3.9. *Every vector space V has a (Hamel) basis.*

Proof. The empty set is a basis of the zero vector space. Suppose that $V \neq 0$, let X be set the set of linearly independent subsets of V ordered by inclusion. First notice that X is nonempty because for any $v_0 \neq 0$ in V we have that $\{v_0\}$ is linearly independent so $\{v_0\} \in X$. Let $\mathcal{C} = \{C_\alpha\}_{\alpha \in I}$ be a chain in X, then take

$$C = \bigcup_{\alpha \in I} C_\alpha$$

We want to show that $C \in X$, i.e. that C is L.I. Take $v_1, v_2, \cdots, v_n \in C$ with

$$r_1 v_1 + r_2 v_2 + \cdots + r_n v_n = 0$$

But $v_i \in C_{\alpha_i}$ for some $\alpha_i \in I$, since C_{α_i} are L.I, then by renumbering we can say

$$C_{\alpha_1} \leq C_{\alpha_2} \leq \cdots \leq C_{\alpha_n}$$

Therefore $v_1, v_2, \cdots, v_n \in C_{\alpha_n}$ hence

$$r_1 = r_2 = \cdots = r_n = 0$$

Hence C is an upper bound of $\mathcal{C}$.

Then by Zorn's lemma X has a maximal element S, so S is linearly independent, we still need to show $\operatorname{Span} S = V$. Suppose it is not, $\exists w \in V \setminus \operatorname{Span} S$, then $S \cup \{w\}$ is linearly independent and $S < S \cup \{w\}$, contradiction. Therefore S is a basis of V. □

Theorem 3.10 (Hahn-Banch)**.** *Let V normed vector space over $\mathbb{R}$ and M supsace of V, and $f \colon M \to \mathbb{R}$ is a bounded linear functional. Then $\exists F \colon V \to \mathbb{R}$ bounded linear functional such that $F_{|M} = f$ and $\|F\| = \|f\|$*

Proof. Let $X = \{(h, D(h))\}$ Where $D(h)$ is a subspace of V containing M and $h\colon D(h) \to \mathbb{R}$ a bounded linear functional, $h_{|M} = f$ and $\|h\| = \|f\|$ with the following partial ordered

$$(h_1, D(h_1)) \leq (h_2, D(h_2)) \iff D(h_1) \subseteq D(h_2) \text{ and } h_{2D(h_1)} = h_1$$

Now clearly $X \neq \varnothing$ since $(f, M) \in X$. Now let $\mathcal{C} = \{(h_\alpha, D(h_\alpha))\}$ be a chain in X. Define

$$D = \bigcup_{\alpha \in I} D(h_\alpha)$$

Then $M \subseteq D$, D is a subspace of V since for$x, y \in D$ and $\lambda \in \mathbb{R}$ we have $x \in D(h_{\alpha_x})$ and $y \in D(h_{\alpha_y})$ but $\mathcal{C}$ is a chain so wlog we have

$$(h_{\alpha_x}, D(h_{\alpha_x})) \leq (h_{\alpha_y}, D(h_{\alpha_y}))$$

Hence $x, y \in D(h_{\alpha_y})$ subspace of V and hence $x + \lambda y \in D(h_{\alpha_y}) \subseteq D$ so D is a subspace of V. Now we define $h\colon D|to\mathbb{R}$ as follows, for $x \in D$, then $x \in D(h_{\alpha_x})$ for some $\alpha_x \in I$, we define

$$h(x) = h_{\alpha_x}(x)$$

- Well defined: If $x \in D(h_{\alpha_1}) \cap D(h_{\alpha_2})$ wlog we have $(h_{\alpha_1}, D(h_{\alpha_1})) \leq (h_{\alpha_2}, D(h_{\alpha_2}))$ which gives us that $h_{\alpha_2}(x) = h_{\alpha_1}(x)$.
- h is linear: this is because each h_α is linear.
- h is bounded: Take $x \in D$ then $x \in D(h_{\alpha_x})$ so

$$|h(x)| \leq \|h_{\alpha_x}\|\|x\| \leq \|f\|\|x\|$$

so h is

todo, Zorn's lemma stuff. □

Lemma 3.11. *V vector space over $\mathbb{C}$ and $f\colon V \to \mathbb{C}$ is linear $\iff$ $f(x) = f_1(x) - if_1(ix)$ with $f_1\colon V \to \mathbb{R}$ $\mathbb{R}$-linear. Moreover, f is bounded iff f_1 is bounded with*

$$\|f\|_* = \|f_1\|_*$$

Proof. ($\implies$) $f\colon V \to \mathbb{C}$, then we can write $f(x) = f_1(x) + if_2(x)$ where $f_1, f_2\colon V \to \mathbb{R}$. Therefore

$$f_1(x) = \frac{f(x) + \overline{f(x)}}{2}$$

We want to show that f_1 is $\mathbb{R}$-linear, so take $\alpha \in \mathbb{R}$ and $x_1, x_2 \in V$ then

$$\begin{aligned} f_1(x+\alpha y) &= \frac{f(x+\alpha y) + \overline{f(x+\alpha y)}}{2} \\ &= \frac{f(x) + \alpha f(y) + + \overline{f(x)} + \overline{\alpha}\overline{f(y)}}{2} \\ &= \frac{f(x) + \overline{f(x)}}{2} + \alpha \frac{f(y) + \overline{f(y)}}{2} \\ &= f_1(x) + f_1(y) \end{aligned}$$

Moreover since f is linear we have that

$$\begin{aligned} f(ix) &= if(x) \\ f_1(ix) + if_2(x) &= if_1(x) - f_2(x) \end{aligned}$$

Therefore $f_2(x) = if_1(ix)$, so

$$f(x) = f_1(x) - if_1(ix)$$

$\Longleftarrow$ Easy.

For the moreover:

$\Longrightarrow$ $|f_1(x)| \leq |f(x)| \leq \|f\|_* \|x\|$ therefore f_1 is bounded with $\|f_1\|_* \leq \|f\|_*$. Now take $x \in V$, so $\exists \Theta_x$ s.t. $f(x) = |f(x)|e^{i\Theta_x}$, therefore

$$\begin{aligned} |f(x)| &= f(x)e^{-i\Theta_x} \\ &= f(e^{-i\Theta_x}x) \\ &= f_1(e^{-i\Theta_x}x) + if_2(e^{i\Theta_x}x) \end{aligned}$$

Comparing real and imaginary parts, we get

$$\begin{aligned} |f(x)| = f_1(e^{-i\Theta_x}x) \\ \leq \|f_1\|_* \|x\| \end{aligned}$$

So

$$\|f\| \leq \|f_1\|$$

$\Longleftarrow$ easy.

□

Theorem 3.12 (Hahn-Banach)**.** *V normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and M subspace of V and $f\colon M \to \mathbb{F}$ bounded linear functional. Then $\exists F\colon V \to \mathbb{F}$ bounded linear functional s.t. $F_{|M} = f$ and $\|F\|_{V^*} = \|f\|_{M^*}$.*

Proof. If $\mathbb{F} = \mathbb{R}$, done. If $\mathbb{F} = \mathbb{C}$, write $f(x) = f_1(x) - if_1(ix)$ with $f_1\colon M \to \mathbb{R}$, so we can extend to $F_1\colon V \to \mathbb{R}$ with

$$\|F_1\| = \|f_1\| = \|f\|$$

Define

$$F(x) = F_1(x) - iF_1(ix)$$

□

Corollary 3.13. *Take $x_0 \in V$ with $x_0 \neq 0$, then $\exists f \in V^*$ s.t. $\|f\| = 1$ and $f(x_0) = \|x_0\|$.*

Proof. Take $M = \operatorname{Span}\{x_0\}$ with $g\colon M \to \mathbb{F}$ defined by

$$g(tx_0) = t\|x_0\|$$

Now take $x \in M$ so $x = tx_0$ then

$$|g(x)| = |t\|x_0\|| = \|tx_0\| = \|x\|$$

Therefore

$$\|g\|_* = 1$$

Apply Hahn-Banach. □

Consequence: If $V \neq \{0\}$, then $V^* \neq \{0\}$.

Corollary 3.14. *$V \neq \{0\}$ normed space, and $x_0 \in V$. Then*

$$\|x_0\| = \sup_{\substack{f \in V^* \\ \|f\|_* 1}} |f(x_0)| = \max_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x_0)|$$

Proof. Let $f \in V^*$ s.t. $\|f\|_* = 1$, then

$$|f(x_0)| \leq \|x_0\| \implies \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x_0)| \leq \|x_0\|$$

But by the corollary above we have that we can attain this sup. □

Corollary 3.15. *$x \neq y$ in V, then $\exists f \in V^*$ s.t. $f(x) \neq f(y)$.*

Proof. $x \neq y$, so $x - y \neq 0$ in V. So by the above, $\exists f \in V^*$ s.t.

$$f(x - y) = \|x - y\| = f(x) - f(y)$$

as desired. □

Corollary 3.16. *V vector space over $\mathbb{F}$ and M proper closed subspace of V, and $x_0 \notin M$ then $\exists f \in V^*$ s.t. $f_{|M} = 0$ and $f(x_0) = 1$.*

Proof. Let $g\colon M \oplus \operatorname{Span}\{x_0\} \to \mathbb{F}$ s.t.

$$g(m + tx_0) = t$$

Then $g(x_0) = 1$ and clearly linear. Moreover take $x = m + tx_0$, with $\|x\| = 1$, then for $t \neq 0$ we have

$$\begin{aligned} \|x\| &= 1 \\ |t|\|m/t + x_0\| &= 1 \\ |t| &\leq \frac{1}{d(x_0, M)} \end{aligned}$$

for $t = 0$ we have $g(m) = 0$, so

$$|g(x)| \leq \frac{1}{d(x_), M}, \quad \forall x \in M \oplus \operatorname{Span}\{x_0\}$$

Then g is bounded, and $g_{|M} = 0$ and $g(x_0) = 1$. We Can extend by Hahn-Banach to some $f \in V^*$. □

Example 3.17. Let $x \in V$ with $f(x) = 0,\ \forall f \in V^*$. Then $x = 0$. This is because

$$\|x\| = \max_{\substack{f \in V^* \\ \|f\|=1}} |f(x)|$$

Now take $\varphi \colon (\ell^1) \to (\ell^\infty)^*$, so

$$\varphi(x) \colon b \to \sum_{n=1}^{\infty} b_n x_n$$

with $b \in \ell^\infty$, then φ is not surjective. This is because C_0 is a closed subspace of ℓ^∞, and $1 = (1, 1, \cdots) \in \ell^\infty \setminus C_0$, then $\exists T \in (\ell^\infty)^*$ s.t. $T_{|C_0} = 0$ and $T(1) = 1$. Suppose that φ is surjective, then $\exists x \in \ell^1$ s.t. $\varphi(x) = T$. So

$$Tb = \sum_{n=1}^{\infty} b_n x_n, \quad \forall b \in \ell^\infty$$

Then since $e_i \in C_0$

$$Te_i = x_i = 0$$

So $x = 0$, hence $T = 0$, contradiction.

Example 3.18. V, W normed space and $V \neq \{0\}$, then $B(V, W)$ is Banach $\iff$ W is Banach.

We already know $\Longleftarrow$. Let w_n be a Cauchy sequence in W. Take $v_0 \in V$ s.t. $\|v_0\| = 1$, then $\exists f \in V^*$ with $\|f\|_* = 1$ and $f(v_0) = 1$. Now $w \in W$, define $T_w \colon V \to W$ given by

$$T_w(x) = f(x) \cdot w$$

Now T_w is linear, and

$$\|T_w x\|_W = |f(x)|\|w\| \leq \|x\|_V \|w\|_W$$

Since this holds for all $x \in V$, we have $T_w \in B(V, W)$ and

$$\|T_w\| \leq \|w\|$$

Moreover,

$$\begin{aligned} \|T_w v_0\| &= |f(x_0)| \|w\| \\ &= \|w\| \end{aligned}$$

with $\|v_0\| = 1$, so

$$\|T_w\| = \|w\|$$

i.e $w \to T_w$ is an isometry. Now since w_n is Cauchy in W, then Tw_n is Cauchy in $B(V, W)$, so Tw_n converges to some T in $B(V, W)$. Take $w = Tv_0$, now notice that

$$\begin{aligned} \|w_n - w\| &= \|f(v_0)w_n - f(v_0)w\| \\ &= \|T_{w_n}(v_0) - T(v_0)\| \\ &\leq \|T_{w_n} - T\| \end{aligned}$$

Therefore $w_n \to w$.

Proposition 3.19. *V normed space, if V^* is separable, then V is separable.*

Remark 3.20. Note that the converse isn't true since ℓ^1 is separable and $(\ell^1)^* = \ell^\infty$ is not.

Corollary 3.21. *$(\ell^\infty)^*$ is not isometrically isomorphic to ℓ^1.*

Proof. V^* is separable, then V^* has a countable dense subset $\{f_n\}$. Now

$$\|f_n\| = \sup_{\substack{x \in V \\ \|x\|=1}} |f_n(x)|$$

So $\exists x_n \in V$ with $\|x_n\| = 1$ and

$$\frac{\|f_n\|_*}{2} \leq |f_n(x_n)| \leq \|f_n\|_*$$

Let $M = \operatorname{Span}\{x_n\}$, we claim that $\overline{M} = V$. Suppose not, then $\exists x_0 \in V$ and $x_0 \notin \overline{M}$, therefore $\exists f \in V^*$ with $f(x_0) = 1$ and $f_{|\overline{M}} = 0$, therefore $f(x_n) = 0 \; \forall n$. We want to show that this implies that $f = 0$. Let $\varepsilon > 0$, and $\exists N$ s.t.

$$\|f_N - f\| < \varepsilon/3$$

So for $\|x\| = 1$,

$$\begin{aligned}
|f(x)| &\leq |f(x) - f(x)| + |f_N(x)| \\
&\leq \|f - f_N\| + \|f_N\| \\
&\leq \|f - f_N\| + 2|f_N(x_N)| \\
&= \|f - f_N\| + 2|f_N(x_N) - f(x_N)| \\
&\leq \|f - f_N\| + 2\|f - f_N\| \\
&< \varepsilon
\end{aligned}$$

Therefore

$$|f(x)| = 0 \implies \|f\| = 0$$

Hence $f = 0$, contradiction.

Now $\overline{\operatorname{Span}\{x_n\}} = V \implies \overline{\operatorname{Span}_{\mathbb{Q}}\{x_n\}} = V$ so V is separable. □

3.2 Adjoint Linear Transformation

Definition 3.22. V_1, V_2 normed space and $T\colon V_1 \to V_2$ a bounded linear transformation. We define the adjoint of T to be a map

$$T^*\colon V_2^* \to V_1^*, \quad (T^* f)(x) = f \circ T$$

Moreover, we can show that T^* is bounded, with

$$\|T^*\| = \|T\|$$

Remark 3.23. In the moreover use Hahn-Banach.

Remark 3.24. We have that

$$(T^* f, x)_{V_1^*, V_1} = (f, Tx)_{V_2^*, V_2}$$

With $T^* \in \mathcal{L}(V_2^*, V_1^*)$, moreover

$$T \in B(V_1, V_2) \iff T^* \in B(V_2^*, V_1^*), \quad \|T^*\| = \|T\|$$

Now

$$(T + \alpha S)^* = T^* + \alpha S^*, \quad \alpha \in \mathbb{F}$$

And

$$(T \circ S)^* = S^* \circ T^*$$

In the homework we prove that T is an isomorphism iff T^* is an isomorphism.

Example 3.25. The Right-Shift operator

$$R\colon \ell^1 \to \ell^1,\ (x_1, x_2, \cdots) \to Rx = (0, x_1, x_2, \cdots)$$

With $\|R\| = 1$, we want to find R^*. We know that

$$R^* \colon (\ell^1)^* \to (\ell^1)^*$$

Take $f \in (\ell^1)^*$, so there exists $b \in \ell^\infty$ s.t.

$$f(x) = \sum_{i=1}^{\infty} b_n x_n := f_b(x)$$

Now

$$\begin{aligned}(R^* f, x)_{(\ell^1)^*, \ell^1} &= (f_b, Rx) \\ &= b_2 x_1 + b_3 x_2 + \cdots \\ &= \sum_{i=1}^{\infty} b_{n+1} x_n\end{aligned}$$

Therefore

$$R^* f_b = f_{(b_2, b_3, \cdots)} = f_{L(b)}$$

Abusing notation we can say that

$$R^* b = Lb$$

3.3 Reflexive Spaces

Definition 3.26. Let V a normed space and V^* be the dual of V, $B(V, \mathbb{F})$ and V^{**} the dual of V^* $B(V^*, \mathbb{F})$. Define the canonical map (the evaluation map)

$$J_V \colon V \to V^{**},\ (J_V x)(f) = f(x)$$

i.e.

$$(J_1 x, f)_{V^{**}, V^*} = (f, x)_{V^*, V}$$

- J_V is linear:

$$\begin{aligned}(J_V(x + \alpha y), f) &= f(x + \alpha y) \\ &= f(x) + \alpha f(y) \\ &= (J_V x, f) + \alpha (J_V y, f) \qquad\qquad = (J_V(x) + \alpha J_V(y), f)\end{aligned}$$

-

$$\begin{aligned}\|J_V x\|_{V^{**}} &= \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |(J_V x, f)| \\ &= \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x)| \\ &= \|x\|\end{aligned}$$

Where the last equality is by Hahn-Banach, so J_V is an isometry. Therefore

$$V \cong J_V(V)$$

Therefore $V^*{*}$ contains an isomorphic copy of V.

Definition 3.27. We say that V is reflexive iff J_V is surjective.

Remark 3.28. All reflexive spaces are Banach.

Example 3.29.

- ℓ^1 is not reflexive, since if it was reflexive, then $(\ell^1)^{**} \cong \ell^1$, but $(\ell^1)^{**} \cong (\ell^\infty)^*$ but this would mean that ℓ^∞ is separable, contradiction.
- ℓ^p is reflexive for $1 < p < \infty$. Let

$$J_p \colon \ell^p \to (\ell^p)^{**}$$

We want to show that this is surjective with

$$\begin{aligned}(J_p x, f) &= (f, x) \\ &= \sum_{n=1}^{\infty} b_n x_n \\ &= \sum_{n=1}^{\infty} x_n b_n \\ &= (f_x, b)_{(\ell^q)^*, \ell^q} \\ &= (J_q b, f_x)_{(\ell^q)^{**}, (\ell^q)^*}\end{aligned}$$

Take $\xi \in (\ell^p)^{**}$, now

$$\xi \colon (\ell^p)^* \to \mathbb{R}$$

and we have some isomorphism $\varphi \colon \ell^q \to (\ell^p)^*$ with $\varphi(b) = f_b \colon x \in \ell^p \to \sum b_n x_n$. Now we have an isomorphism

$$\varphi^* \colon (\ell^p)^{**} \to (\ell^q)^*$$

Now

$$\varphi^*(\xi) \in (\ell^q)^*$$

so $\exists x \in \ell^p$ s.t. $\varphi^*(\xi) \colon b \in \ell^q \to \sum b_n x_n$. Now we need to show that

$$J_p(x) = \xi$$

Indeed, for $f \in (\ell^p)^*$ we have

$$\begin{aligned}(\xi, f) &= (\xi, \varphi(b)) \\ &= (\varphi^* \xi, b) \\ &= \sum b_n x_n \\ &= \sum x_n b_n \\ &= f_b(x) \\ &= (J_V x, f_b) \\ &= (J_V x, f)\end{aligned}$$

Lemma 3.30. *Let V_1, V_2 be normed vector spaces and let $T : V_1 \to V_2$ be a bounded linear operator. Then the following diagram commutes:*

$$\begin{array}{ccc} V_1 & \xrightarrow{T} & V_2 \\ {\scriptstyle J_{V_1}}\downarrow & & \downarrow{\scriptstyle J_{V_2}} \\ V_1^{**} & \xrightarrow[T^{**}]{} & V_2^{**} \end{array}$$

That is,

$$T^{**} \circ J_{V_1} = J_{V_2} \circ T.$$

Moreover, if T is a bounded linear isomorphism with bounded inverse, then T^* is also an isomorphism and

$$(T^*)^{-1} = (T^{-1})^*.$$

Proof. Let $x \in V_1$ and $g \in V_2^*$, then

$$\begin{aligned}(T^{**} \circ J_{V_1} x, g)_{V_2^{**}, V_2^*} &= (J_{V_1} x, T^* g)_{V_1^{**}, V_1^*} \\ &= (T^* g, x)_{V_1^*, V_1} \\ &= (g, Tx)_{V_2^*, V_2} \\ &= (J_{V_2}(Tx), g)_{V_2^{**}, V_2^*}\end{aligned}$$

Therefore

$$(T^{**} \circ J_{V_1})(x) = (J_{V_2} \circ T)(x)$$

Since this holds for every $x \in V_1$, we get the desired result. □

Corollary 3.31. *If $V_1 \cong V_2$, then V_1 is reflexive $\iff$ V_2 is reflexive.*

Lemma 3.32. *A closed subspace of a reflexive space is reflexive.*

Proof. Let V be a reflexive space, and M closed subspace of V - wlog we take M to be proper. Consider the canonical embedding

$$J_V : V \to V^{**}, \quad J_M : M \to M^{**}$$

and we know that J_V is surjective, we want to show that J_M is surjective. Let $\xi \in M^{**}$, i.e. $\xi \colon M^* \to \mathbb{F}$. Define $\eta \colon V^* \to \mathbb{F}$ a linear functional with

$$\eta(f) = \xi(f_{|M})$$

Note that

$$\begin{aligned} |\xi(f)| &= |\xi(f_{|M})| \\ &\leq \|\xi\| \|f_{|M}\| \\ &\leq \|\xi\| \|f\| \end{aligned}$$

Hence $\eta \in V^{**}$ (with $\|\eta\| = \|\xi\|$). Therefore since J_V is surjective, $\exists x_0 \in V$ with $J_V x_0 = \eta$. We claim that $x_0 \in M$. Assume not, then since M is closed $\exists f \in V^{**}$ with $f_{|M} = 0, \quad f(x_0) = 1$. But

$$\begin{aligned} (J_V x_0, f) &= f(x_0) \\ (\eta, f) &= 1 \\ (\xi, f_{|M}) &= 1 \\ 0 &= 1 \end{aligned}$$

Contradiction. We still need to show that $J_M(x_0) = \xi$. Let $h \in M^*$, let H be a Hahn-Banach extension to V^*.

$$\begin{aligned} (J_M x_0, h) &= (J_V x_0, H) \\ &= (\eta, H) \\ &= (\xi, H_{|M}) \\ &= (\xi, h) \end{aligned}$$

□

Lemma 3.33. *Let V be a normed space, and consider the canonical maps*

$$J_V \colon V \to V^{**}, \qquad J_V^* \colon V^{***} \to V^*, \qquad J_{V^*} \colon V^* \to V^{***}.$$

Then the following diagram commutes:

$$\begin{array}{ccc} V^* & \xrightarrow{J_{V^*}} & V^{***} \\ & \searrow^{\mathrm{id}_{V^*}} & \downarrow J_V^* \\ & & V^* \end{array}$$

That is,

$$J_V^* \circ J_{V^*} = \mathrm{id}_{V^*}.$$

In particular, J_V^* is always surjective.

Proof. Let $f \in V^*$, let $\xi := J_{V^*} f \in V^{***}$.

$$\begin{aligned}((J_V^* \circ J_{V^*})f, x)_{V^*,V} &= (J_{V^*} f, J_V^* x)_{V^*,V^{**}} \\ &= (J_V x, f) \\ &= (f, x)\end{aligned}$$

Therefore

$$J_V^* \circ J_{V^*}(f) = f$$

as desired. □

Theorem 3.34. *V reflexive $\iff$ V Banach and V^* reflexive.*

Corollary 3.35. *We have that $\ell^\infty \cong (\ell^1)^*$, if ℓ^∞ is reflexive then $(\ell^1)^*$ is reflexive with ℓ^1 Banach. Therefore ℓ^1 is reflexive, contradiction.*

Proof. $\implies$ V reflexive then J_V is an isometric isomorphism, therefore so is J_V^*. From the last lemma we get that J_{V^*} is also an isometric isomorphism. So V^* is reflexive. We already know that V is Banach by reflexive.

$\impliedby$ Let V Banach with V^* reflexive. Then by the above V^{**} is reflexive, but we have that $V \cong J(V)$ subspace of V^{**}. It is enough to show that $J(V)$ is closed. But V is Banach and J is an isometry, therefore $J(V)$ is also Banach and hence closed.

□

4 Pillars of Functional Analysis

4.1 Uniform Boundedness Principle.

Theorem 4.1 (Uniform Boundedness Principle)**.** *V is a Banach space and W a normed space. Now let $\mathcal{T}_{\alpha\alpha\in I}$ a family of bounded linear transformations $V \to W$ that are pointwise bounded, i.e.*

$$\sup_{\alpha\in I}\|T_\alpha x\| < \infty$$

Then $\{T_\alpha\}$ is uniformly bounded, that is $\exists M > 0$ s.t.

$$\|T_\alpha\|_* < M, \quad \forall\alpha \in I$$

In other words,

$$\|T_\alpha x\| < M\|x\|, \quad \forall x \in V,\ \forall\alpha \in I$$

Proof. For every $n \in \mathbb{N}$, let $E_n = \{x \in V \colon \|T_\alpha x\| \leq n\ \forall\alpha \in I\}$, clearly E_n is closed since we can write

$$E_n = \bigcap_{\alpha\in I} \{x \in V \colon \|T_\alpha x\|\} \leq n$$

Now for every $x \in V$, we have $\sup_{\alpha\in I}\|T_\alpha x\| < \infty$ so there exists $N := N_x$ s.t.

$$\|T_\alpha x\| \leq N_x, \quad \forall x \in V$$

Therefore $x \in E_{N_x}$. Hence

$$V = \bigcup_{n=1}^{\infty} E_n$$

With V Banach. Then by BCT, $\exists n_0$ s.t. E_{n_0} contains nonempty interior, i.e. $\exists x_0 \in V$ and $r_0 > 0$ s.t.

$$B(x_0, r_0) \subseteq E_{n_0}$$

Now let $x \in V$ s.t. $\|x\| = 1$, take $y = x_0 + \frac{r_0}{2}x$ then

$$y \in B(x_0, r_0) \subseteq E_{n_0}$$

We can write

$$x = \frac{2}{r_0}(y - x_0)$$

So $\forall x$ with $\|x\| = 1$ we have

$$\begin{aligned} \|T_\alpha x\| &= \frac{2}{r_0}\|T_\alpha y - T_\alpha x_0\| \\ &\leq \frac{2}{r_0}\left(\|T_\alpha y\| + \|T_\alpha x_0\|\right) \\ &\leq \frac{4n_0}{r_0} \end{aligned}$$

Hence we get

$$\|T_\alpha\| \leq \frac{4n_0}{r_0}, \quad \forall \alpha \in I$$

□

Corollary 4.2. *V Banach, W normed space and T_n a sequence in $B(V, W)$ that converges pointwise to some T.*

i) T_n are uniformly bounded.

ii) $T \in B(V, W)$ with $\|T\| \leq \liminf_{n\to\infty}\|T_n\|$

Proof. Take $x \in V$, since $T_n x$ converges in W it is bounded. Therefore by the uniform boundedness principle we have that T_n is uniformly bounded. Now we have that

$$T_n x \leq M\|x\|$$

Let $n \to \infty$, we get that $T \in B(V, W)$. Moreover, we know that

$$\|T_n\|x \leq \|T_n\| \cdot \|x\|$$

Therefore

$$\|T_n\|x = \liminf\|T_n\|x \leq \liminf\|T_n\|\|x\|$$

□

Example 4.3. V normed space and $A \subseteq V$ s.t. $\{f(x)\colon x \in A\}$ is bounded in $\mathbb{F}$ $\forall f \in V^*$. then A is bounded in V^*. For every $x \in A$ we define $T_x\colon V^* \to \mathbb{F}$ by $T_x f = f(x)$. Now $\{T_x\}_{x\in A}$ is pointwise bounded s.t. for every $f \in V^*$ we have

$$\|T_x f\| = \|f(x)\| \leq M_f$$

With V^* Banach, therefore $\{T_x\}$ is uniformly bounded. Therefore $\exists M > 0$ s.t. $\forall x \in A$

$$\begin{aligned} \|T_x\| &\leq M \\ \sup_{\substack{f\in V^* \\ \|f\|_*=1}} \|T_x f\| &\leq M \\ \sup_{\substack{f\in V^* \\ \|f\|_*=1}} &\leq M \\ \|x\| &\leq M \end{aligned}$$

Since this holds for every $x \in A$ we get that A is bounded.

Example 4.4. Let $1 < p < \infty$, given a sequence $x = (x_1, x_2, \cdots)$ s.t. $\sum_{n=1}^{\infty} x_n y_n$ converges for all $y \in \ell^p$. Then $x \in \ell^q$ where $\frac{1}{p} + \frac{1}{q} = 1$. Take $T_N\colon \ell^p \to \mathbb{F}$ given by

$$T_N y = \sum_{n=1}^{N} x_n y_n$$

We know that T_N is bounded since

$$\begin{aligned} |T_N y| &= |\sum_{n=1}^{N} x_n y_n| \\ &\leq \left(\sum_{n=1}^{N} |x_n|^q\right)^{1/q} \|y\|_p \end{aligned}$$

Therefore $T_N \in (\ell^p)^*$ by the given we have $T_N \to T(\ell^p)^*$. Then $\exists b \in \ell^q$ s.t.

$$Ty = \sum_{n=1}^{\infty} b_n y_n = \sum_{n=1}^{\infty} x_n y_n$$

Take $y = e_i$, $b_i = x_i$.

Example 4.5. V is Banach, and $A \subseteq V^*$ s.t. $\forall x \in V$ we have $\{f(x)\colon f \in A\}$ is bounded, then A is bounded in V^*. For every $f \in A$, define the map $T_f\colon V \to V^*$ defined by

$$T_f(x) = f(x)$$

Now $\{T_f\}_f$ is pointwise bounded with V Banach so it is uniformly bounded. Therefore $\{T_f\}$ is bounded in V^*.

4.2 The Open Mapping Theorem

Remark 4.6.
- $T(A+\alpha B) = T(A) + \alpha T(B)$.
- $\overline{\alpha A} = \alpha \overline{A}$.
- $\overline{A} + \overline{B} \subseteq \overline{A+B}$.
- $B(x,R) = x + RB(0,1)$.

Theorem 4.7 (Open Mapping Theorem)**.** *Let V and W Banach and $T \in B(V,W)$ surjective. Then 0_W is an interior point of $T(B_V(0,1))$. Consequently T is an open map.*

Proof. **Step 1.** $\exists r > 0$ s.t. $B_W(0,r) \subseteq \overline{T(B_V(0,1))}$. Now T is surjective, so we can write

$$W = T(V) = \bigcup_{n=1}^{\infty} \overline{T(B_V(0,n))}$$

Now $\overline{T(B_V(0,n))}$ are closed in W, a Banach space. Therefore by BCT, $\exists n_0$ s.t. $\overline{T(B_V(0,n_0))}$ has nonempty interior. But

$$\overline{T(B_V(0,n_0))} = n_0 \overline{T(B_V(0,1))}$$

Which contains an open ball. Now $\overline{T(B_V(0,1))}$ contains a ball $B_W(y,R)$. So we have that

$$\begin{aligned} B(y,R) &\subseteq \overline{T(B_V(0,1))} \\ y + B(0,R) &\subseteq \overline{T(B_V(0,1))} \\ B(0,R) &\subseteq \overline{T(B_V(0,1))} - y \\ &\subseteq \overline{T(B_V(0,1))} + \overline{T(B_V(0,1))} \\ &\subseteq \overline{T(B_V(0,2))} \\ &= 2\overline{T(B_V(0,1))} \end{aligned}$$

Therefore

$$\frac{1}{2} B(0,R) \subseteq \overline{T(B_V(0,1))}$$

Step 2. If $B_W(0,R) \subseteq \overline{T(B_V(0,1))}$ then $B_W(0,r) \subseteq T(B_V(0,2))$. Therefore

$$B_W(0,r/2) \subseteq T(B_V(0,1))$$

i.e. 0_W is an interior point of $T(B_V(0,1))$.
We have that $B_W(0,r) \subseteq \overline{T(B_V(0,1))}$, take $y \in B_W(0,r)$. Therefore $y \in \overline{T(B_V(0,1))}$, so $\exists x_1 \in B_V(0,1)$ ($\|x_1\| < 1$) s.t. $\|y - Tx_1\|_W < r/2$. Therefore $y - Tx_1 \in B(0,r/2) \subseteq \overline{T(B_V(0,1/2))}$, then $\exists x_2$ s.t. $\|x_2\| < 1/2$ and $\|(y - Tx_1) - Tx_2\| < r/2^2$. Recursively, we construct a sequence x_n s.t. $\|x_n\| < 1/2^{n-1}$ and $\|y - (Tx_1 + \cdots + Tx_n)\| < r/2^n$. But $\sum \|x_n\| < \infty$ therefore since V

is Banach we have that $\sum x_n$ converges to some $x \in V$. Then by continuity of T we have that

$$\begin{aligned} T(x) &= T\left(\sum x_n\right) \\ &= \sum T(x_n) \\ &= y \end{aligned}$$

But we have that

$$\begin{aligned} \|x\| &\leq \sum_{n=1}^{\infty} \|x_n\| \\ &< \sum \frac{1}{2^{n-1}} \\ &= 2 \end{aligned}$$

Then $y \in T(B_V(0,2))$.

Step 3. T is an open map. Let U be open in V, we want to show that $T(U)$ is open in W. Take $y \in T(U)$, then $\exists x \in U$ s.t. $y = Tx$ but U is open so $\exists r > 0$ with

$$B(x,r) \subseteq U$$

Therefore

$$y \in T(B(x,r)) \subseteq T(U)$$

But we have that

$$\begin{aligned} B(x,r) &= x + B(0,r) \\ &= x + rB(0,1) \end{aligned}$$

So

$$TB(x,r) = y + rT(B(0,1))$$

Now 0_W is an interior point of $T(B(0,1))$, then $\exists c > 0$ s.t. $B(0,c) \subseteq T(B(0,1))$. Then

$$B_W(y,rC) = y + rB_W(0,c) \subseteq y + rT(B(0,1)) \subseteq T(U)$$

□

Corollary 4.8 (Inverse function Theorem)**.** *Let V, W Banach and $T \in B(V,W)$ bijective then $T^{-1} \in B(W,V)$.*

Proof. T is open, then T^{-1} continuous. □

Corollary 4.9. *Let $\|\cdot\|_a,\ \|\cdot\|_b$ norms on V s.t. $(V, \|\cdot\|_a)$ and $(V, \|\cdot\|_b)$ are Banach and $\exists c$ s.t.*

$$\|x\|_b \leq c\|x\|_a$$

Then the norms are equivalent.

Proof. Let $\mathrm{id}\colon : V_a \to V_b$, this is continuous so $\mathrm{id}_V \in B(V_a, V_b)$ and bijective - so its inverse is also continuous. Therefore $\exists M$ s.t.

$$\|x\|_a = M\|x\|_b$$

□

4.3 The Closed Graph Theorem

Definition 4.10. Let (X, τ_x) and (Y, τ_y) topological spaces, and $f\colon X \to Y$. Then

$$\mathcal{G}(f) = \{(x, f(x))\colon x \in X\} \subseteq (X \times Y, \tau_x \times \tau_y)$$

Proposition 4.11. *If Y is Hausdorff and f is continuous, then $\mathcal{G}(f)$ is closed.*

Proof. Take $(x, y) \in \mathcal{G}(f)^c$, then $y \neq f(x)$ and Y is Hausdorff. Therefore $\exists V_y, V_{f(x)}$ open in Y s.t.

$$V_y \cap V_{f(x)} = \varnothing$$

But f is continuous, therefore $\exists U_x \ni x$ open s.t.

$$f(U_x) \subseteq V_{f(x)}$$

Hence

$$U_x \times V_y \subseteq \mathcal{G}(f)^c$$

as desired. □

Remark 4.12. In the particular case of normed spaces we have $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ normed spaces with $(V \times W, \|\cdot\|_{V\times W})$ define as follows:

$$\|(v, w)\| = \|v\| + \|w\|$$

1. If V and W are Banach then $V \times W$ is also Banach.
2. If $T\colon V \to W$ in $L(V, W)$, then $\mathcal{G}(T)$ is a subspace of $V \times W$.

Example 4.13. Take $\frac{d}{dx}\colon (C^1([0,1]), \|\cdot\|_\infty) \to (C([0,1]), \|\cdot\|_\infty)$. Then

$$\mathcal{G}(\frac{d}{dx}) = \left\{(f, f')\colon f \in C^1([0,1])\right\}$$

Now take $(f_n, f_n') \to (f, g)$ with $f \in C^1([0,1])$, $g_n \in C([0,1])$, then $f_n \to f$ and $f_n' \to g$ uniformly, then $f' = g$. Therefore $(f, g) \in \mathcal{G}(\frac{d}{dx})$ but $\frac{d}{dx}$ is not continuous.

Theorem 4.14 (Closed Graph Theorem)**.** *V, W Banach spaces with $T \in \mathcal{L}(V, W)$, then $T \in B(V, W) \iff \mathcal{G}(T)$ is closed in $V \times W$.*

Proof($\Longrightarrow$) Always true since W is a metric space (so Hausdorff).
($\Longleftarrow$) $\mathcal{G}(T)$ is closed in $V \times W$, which is Banach - therefore $\mathcal{G}(T)$ is Banach. We define

$$\pi_1 \colon \mathcal{G}(T) \to V, \quad \pi_1(x, Tx) = x$$

Clearly this is a bijective L.T with

$$\|\pi_1(x, Tx)\| = \|x\| \leq \|(x, Tx)\|$$

Therefore π_1 is bounded, so by IFT we get that π^{-1} is bounded. Then $\exists M > 0$ s.t.

$$\|Tx\| \leq \|(x, Tx)\| = \|\pi_1^{-1}(x)\|_{V \times W} \leq M\|x\|$$

Therefore T is bounded.

□

Example 4.15. Let V and W Banach, and $T \in \mathcal{L}(V, W)$ such that for every $S \in W^*$ we have $S \circ T \in V^*$ therefore $T \in B(V, W)$. We prove that $\mathcal{G}(T)$ is closed in $V \times W$, then apply the Closed graph theorem. Indeed, take a sequence (x_n, Tx_n) converging $(x, y) \in V \times Y$. We want to show that $y = Tx$. Indeed, we have that $\forall S \in W^*$

$$\lim_{n\to\infty} S \circ T(x_n) = S \circ T(x) = \lim_{n\to\infty} S(Tx_n) = Sy$$

Therefore $S(Tx) = Sy$ and hence

$$S(y - Tx) = 0, \quad \forall S \in W^*$$

By Hahn-Banach, we get that

$$y - Tx = 0$$